

In Awe of Planning: An Exploration of the Role of Awe in the Built Environment

By

Tyler Barron

Bachelor of Science in Sustainable Food and Farming
Bachelor of Arts in Community Collaboration and Sustainable Together-Living
University of Massachusetts, Amherst

Submitted to the Department of Urban Studies and Planning in partial fulfillment of the
requirements for the degree of

MASTER IN CITY PLANNING

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

May 2026

© Tyler Barron. All Rights Reserved.

The author hereby grants to MIT a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by: Tyler Barron
Department of Urban Studies and Planning
April 28, 2026

Certified by: Ezra Glenn
Lecturer in Urban Studies and Planning
Department of Urban Studies and Planning, Thesis supervisor

Accepted by: J. Phillip Thompson
Professor of Political Science and Urban Planning
Department of Urban Studies and Planning, MCP Committee Chair

In Awe of Planning: An Exploration the Role of Awe in the Built Environment

by

Tyler Barron

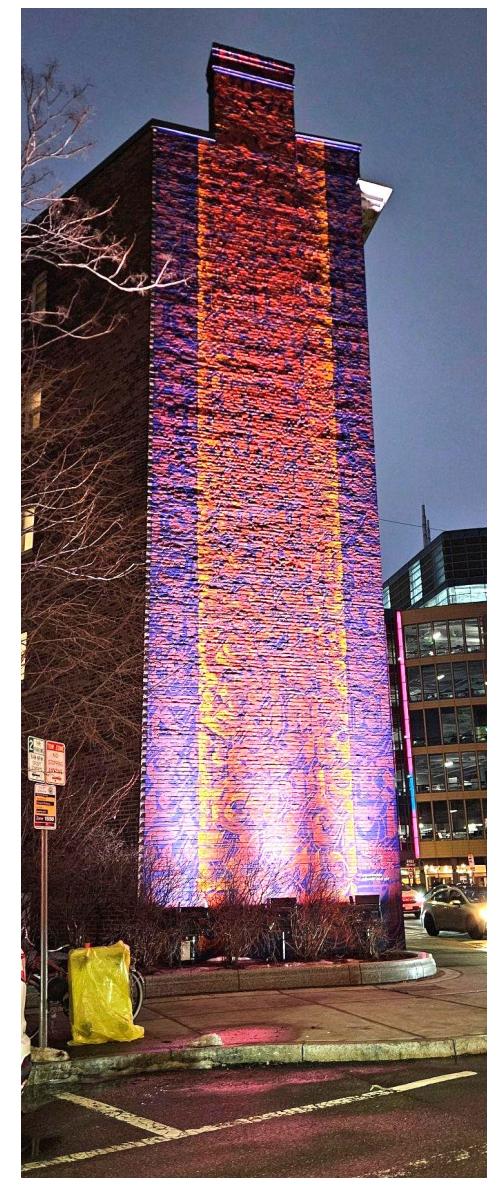
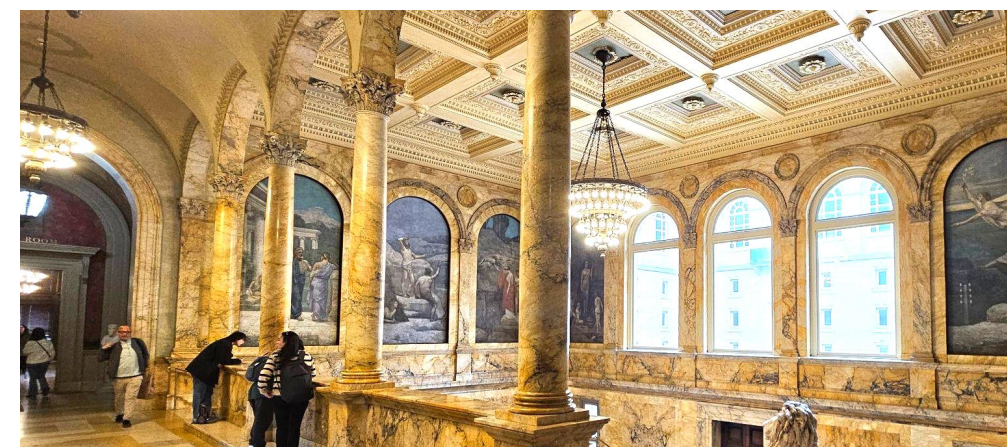
Submitted to the Department of Urban Studies and Planning on April 28, 2026 in
Partial Fulfillment of the Requirements for the Degree of Master in City Planning

ABSTRACT

A growing body of research on awe has demonstrated its capacity to promote prosocial behavior, improve well-being, and foster a sense of connection with larger systems. However, its relationship to the built environment remains largely unexplored. Urban planners and designers have few frameworks for understanding where awe occurs in cities, what elicits awe, and how it might be intentionally cultivated. In partnership with Gehl's Cities of Awe Lab, this research uses Cambridge, Massachusetts as the primary site of inquiry to examine how people experience awe in everyday public spaces. In order to interpret awe—which is thought to arise in the presence of something vast that transcends one's current understanding of the world—this research develops an original analytical framework. Organized across geographies, morphologies, phenomenologies, and temporalities of awe, and informed by Dacher Keltner's science of awe, Henri Lefebvre's Rhythmanalysis, and theories of sacredness, this framework yields insights on the nature of awe in Cambridge. Drawing on interviews with two participant groups, awe walk interventions, and personal observation, which together provided a collection of 220 georeferenced instances of awe, four archetypes of awe emerge from the quantitative analysis: Expansive Nature, Wild Nature, Built Beauty, and Inclusive Interaction. This analysis further reveals that street-based and park-based settings are the most common locations of awe, while plants, architecture, and art and objects are the most frequent elicitors of awe. Participants affiliated with nature-related activities tended towards ecologically and socially driven awe, while those without such affiliations more commonly experienced aesthetically and socially driven awe. Awe walk interventions produced consistent benefits across both groups, including heightened sensory awareness, increased gratitude, and stronger connection to place. As interest in the affective and psychological dimensions of city life grows, this research offers planners and designers a practical framework—anchored in an interactive Awe Atlas—for identifying, preserving, and cultivating the conditions that offer a sense of vastness and transcendence.

Thesis supervisor: Ezra Glenn

Title: Lecturer in Urban Studies and Planning



In Awe of Planning: An Exploration of the Role of Awe in the Built Environment

By Tyler Barron



Table of Contents

1. Introduction	
1.1 Project Overview	3
1.2 Acknowledgements	4
2. Conceptual Foundations for Awe	
2.1 The Science of Awe	7
2.2 Rhythmanalysis	10
2.3 Sacredness and Awe	13
3. Methodology	
3.1 Research Design and Data Collection	19
3.2 Analytical Framework for Awe	22
3.3 Quantitative Methods	33
4. Cantabrigian Awe	
4.1 Qualitative Findings	37
4.2 Quantitative Findings	45
4.3 Awe by Affiliation	65
4.4 An Awe Profile of Cambridge	
4.5 Limitations	
5. Implications for Planning	
5.1 Cities of Awe	69
5.2 Finding Rhythm	73
5.3 Activating Awe	79
5.4 The Awe Atlas	92
References & Appendices	95

Section One



Introduction

Awe occurs in a realm separate from the mundane world of materialism, money, acquisition, and status signaling—a realm beyond the profane that may be called the sacred.

- Dacher Keltner, *Awe*, 2023, p. 17

The sacred experience involves the person in a new conceptual world in which he: abandons his fixed world momentarily; attaches new meaning through direct experience to an object or place; reorders his perceptions according to the insights provided by this new framework.

- Karalyn Emilie Krasin, *Sacred Experience and Place in Cities*, 1972, p. 66

I.1 Project Overview

This investigation into the role of awe in the built environment comes out of a semester-long Master in City Planning thesis at the Massachusetts Institute of Technology. As a client-based thesis, the project was conducted with the support of an external partner, Gehl. More specifically, this work reflects the research interests of the Cities of Awe Lab, a collaborative initiative between Gehl, the University of California, Berkeley, and The Gambrell Foundation. The Lab's purpose is to use science, design, and imagination as tools to transform urban environments into places that promote connection, joy, awe. The Lab's efforts are geared toward understanding where and how awe appears in cities, how it can be nurtured, and how it influences people and communities over time (Gehl, n.d.). Each of the Lab's partners offers a distinct strength: Gehl contributes decades of experience in human-centered urban design and strategy; UC Berkeley contributes scientific research on awe, emotion, and social connection through the Greater Good Science Center and the Social Interaction Lab, while The Gambrell Foundation contributes a philanthropic lens that reimagines funding as a catalyst for thriving communities. The Lab's work centers on a set of core questions: How do moments of awe shape public life? When does public life support experiences of awe? What types of shared spaces and environments make these experiences possible?

In an effort to align with research interests of the Cities of Awe Lab, this thesis takes up these questions in the context of a single city—Cambridge, Massachusetts—and develops an analytical framework for understanding how experiences of awe might be examined elsewhere. To this end, the primary research questions of this study focus on the differences in how people experience awe and on the public spaces and urban design elements most closely associated with this emotional state:

- How do people experience awe in everyday public spaces in Cambridge, and how does this differ between groups oriented toward awe-related experiences and groups not explicitly oriented toward awe?
- What kinds of public spaces are most frequently associated with awe?
- What urban design elements are most closely associated with participants' experiences of awe?
- How do awe interventions shape perception of public space over time?

I.2 Acknowledgements

I am grateful for the many people who supported this project and who helped me to navigate the often groundless experience of understanding awe. Thank you to Olivia Flynn and Louise Vogel Kielgast from Gehl for helping to guide this research, for welcoming me to Copenhagen, and providing this opportunity to be changed by the work of thinking about and feeling through awe. Thank you also to Adriana Akers from Gehl for generously responding to my cold email at the start of the MCP program, and for being willing to support my interest in a client-based thesis nearly a year later.

Thank you to each of the 11 participants who took part in this study and who were willing to engage with such an unusual research topic. I deeply appreciate your time, thoughtfulness, and willingness to go on awe walks. I found our conversations to be incredibly insightful and rewarding and I sincerely hope that I have represented your sentiments and experiences as accurately as possible.

Thank you to Ezra Glenn for advising this work, and thank you to Brent Ryan and Mariana Arcaya for offering fresh perspectives on this thesis process. I am also grateful to my fellow classmates for the past two years of learning and comradery. In particular, thank you Riccardo Fiorista for helping me to get a handle on this project at the last moment, and thank you to Takumi Kitamura for your ongoing support, curiosity, and care as both a classmate and a friend.

To Jordan, my partner, thank you for enduring this journey with me. Thank you for the countless supportive and insightful conversations as I tried to make sense of this process and thank you for your generosity, love and kindness as I fell down this academic rabbit hole.

AI Usage Disclosure:

This thesis, *In Awe of Planning: An Exploration of the Role of Awe in the Built Environment*, was developed with assistance from a large language model, Claude (Anthropic, 2026). AI was used to support the identification and implementation of statistical techniques for awe-based categorical data, including the generation and debugging of code in R. AI was also used to help organize, synthesize, and edit research notes and written text and refine the structure of certain chapters. AI was not used to develop the central arguments of this thesis nor was it used to make analytical or interpretive decisions. All conceptual frameworks, findings, and interpretations are the author's own. For further information on the extent and nature of AI usage, please contact the author.

Section Two



Conceptual Foundations for Awe

Our sensations and perceptions, in full and continuous appearances, contain repetitive figures, concealing them.

Thus, sounds, lights, colours and objects. We contain ourselves by concealing the diversity of our rhythms: to ourselves, body and flesh, we are almost objects. Not completely, however. But what does a midge perceive, whose body has almost nothing in common with ours, and whose wings beat to the rhythm of a thousand times per second? This insect makes us hear a high-pitched sound, we perceive a threatening, little winged cloud that seeks our blood. In short, rhythms escape logic, and nevertheless contain a logic.

- Henri Lefebvre, *Rhythmanalysis*, 2004 (p. 10)

2.1 The Science of Awe

Awe and its Effects:

Awe is a complex emotion defined as a “the feeling of being in the presence of something vast that transcends your current understanding of the world” (Keltner, 2023, p. 7). Perception of awe can have positive or negative valence (i.e. something which is awe inspiring vs. something that is awful) and maybe be brief and mild, or profound, memorable and intense (Büssing, 2021). Awe can be triggered by both internal and external stimuli and it has been found that awe is experienced an average of two to three times per week (Keltner, 2023, p. 26). From a physiological standpoint, the feeling of awe commonly involves a vocal burst, “a pattern of sound that lasts a quart of second... [that] is intended to convey emotion” (Keltner, 2023, p. 57), in the form of a ‘woah’ or ‘wow.’ It can also be accompanied by smaller physical shifts like a slightly upward tilted head, a smile, a drop-jawed mouth, widened eyes, raised inner eyebrow, and an inhalation (Allen, 2018, p. 11). More dramatic responses include tears which are understood to “arise in recognition of those larger patterns that unite us,” and chills which “signal to us that we are seeking to make sense of such unknowns with others” (Keltner, 2023, p. 65). Beyond these short-term physiological effects, repeated experiences of awe have been found to have lasting positive impacts, including “elevated vagal tone, reduced sympathetic activation, increased oxytocin, and reduced inflammation” (Monroy & Keltner, 2023, p. 311). One study in particular found that when compared to a six other positive emotions, awe was the only one negatively associated with proinflammatory cytokines—proteins that have been linked to a number of chronic diseases (Stellar et al., 2015).

From a psychological standpoint, awe has been shown to have a range of effects including a reduced sense of self, prosocial behavior, and the need for accommodation. Monroy & Keltner (2023) point out that “across diverse methodologies—lab studies, daily diaries, online narratives, in vivo nature studies—and elicitors of awe, such as images of nature or visual art, awe diminishes self-focus” (p. 311). Though this may sound like a potentially undesirable outcome, a metaphorically diminished sense of subjective or physical sense, also termed small-self, can lead to less attention being focused on day-to-day concerns and more attention being given to things larger than oneself like community, culture, or nature (Piff et al., 2015, p. 884). Importantly, this “shift in attention toward larger entities and diminishment of the individual self [is one] that is vital to the collaboration and cooperation required of social groups” (p. 884). Awe’s ability to invoke a small-self is thought to mediate expressions of humility (Yaden et al., 2019) as well as tendencies towards collective engagement (Bai et al., 2017), cooperation, sacrifice, sharing (Monroy & Keltner, 2023, p. 312), empathy and gratitude (Piff et al., 2015, p. 885). Piff et al. (2015) argue that “to the extent that awe triggers feelings that one’s being and goals are less significant relative to something vaster than the self, it should also increase prosocial tendencies” (p. 885). Such prosocial tendencies have themselves been found to elevate well-being and even boost life expectancy (Monroy & Keltner, 2023, p. 312). Keltner (2023, p. 37) highlights this point by explaining, “as our default self vanishes...awe shifts us from a competitive, dog-eat-dog mindset to perceive that we are part of networks of more interdependent, collaborating

individuals.” The potential for awe to inspire interdependence and connection has been further reinforced by studies indicating that awe helps people to feel more connected to other people, to humanity, and to things around them. (Allen, 2018, p. 30).

In addition to a small sense of self and prosocial behavior, an key effect of awe is the need for accommodation. Similar to small-self, the need for accommodation can impact self-perception as it involves “changes to existing mental schemas in order to mentally process and integrate an experience” (Yaden et al., 2019, p. 2). Accommodation, in the context of awe, typically arises from encountering stimuli that exceed our expectation or normal experience and which therefore provoke a shift, expansion, or reconsideration of our current understanding of the world. It has been theorized that awe’s ability to induce cognitive accommodation “could help people adapt to new environments and experiences by allowing them to process novel, complex information more critically” (Allen, 2018, p. 15). While not directly connected to accommodation, other relevant effects of awe include its ability to “enhance mental and physical health through elevating the individual’s sense of meaning” (Monroy & Keltner, 2023, p. 312), improve people’s mood and sense of well-being, increase people’s sense of life satisfaction (Allen, 2018, pp. 31-32), dampen feelings of materialism (Rudd et al., 2012), and decrease people’s desire for money (Jiang et al., 2018).

Awe and Time:

An intriguing aspect of awe its influence on people’s perception of time. Grounded in the understanding that one’s sense time availability is often an indicator of well-being and a contributor to prosocial behavior, and based on the theory that focusing on the present moment elongates time perception, Rudd et al. (2012) looked at the effects of time across three experiments. The first experiment demonstrated that relative to happiness, awe “led to a perception that time is more plentiful and expansive” (p. 1132). The second experiment in this study indicated that awe’s ability to expand time “led to reduced impatience and increased willingness to volunteer one’s time” (p. 1134), while the third experiment revealed that “eliciting a feeling of awe, compared with a neutral state, increased perceived time availability” (p. 1135). In summary, Rudd et al. (2012) indicate that inducing experiences of awe might serve as a way “of alleviating the feeling of time starvation that plagues so many people in modern life” (p. 1135).

Elicitors of Awe:

Qualitative research by Dacher Keltner, relying on 2,600 narratives across 26 languages, reveals a taxonomy through which to classify awe known as the ‘eight wonders of life.’ The discovery of these ‘wonders’ indicate that across cultures, awe is consistently, though not exclusively, elicited by moral beauty (exceptional virtue courage, kindness, strength, perseverance, character and ability displayed by others), collective effervescence (collective, often synchronous, movement, celebration, or rituals), nature (immersion in and appreciation of natural systems), music (rhythms and harmonies that offer symbolic meaning and emotional resonance), visual design (art, architecture, and creativity that express culture, imagination, and

Conceptual Foundations for Awe

beauty), spirituality/religion (encounters with sacred, mystical, transcendent, or mysterious), life & death (moments of appreciation for the cycle of life), and epiphany (big ideas, discoveries or personal realizations about life's essential truths).

Additionally, Keltner & Haidt (2003) identify five themes related to the emotional experience of awe. These 'flavor of awe' include: threat (a feeling of danger or fear), beauty (aesthetic pleasure), ability (exceptional talent or skill), virtue (strength of character and human goodness), and supernatural causality (the perception of some supernatural entity or occurrence) (Keltner & Haidt, 2003, pp. 305-306). Regardless of the flavor or wonder, scientific research on awe postulates that "experiences of awe are unified by a core theme: perceptions of vastness that dramatically expand the observer's usual frame of reference in some dimension or domain (Piff et al., 2015, p. 884). The perception of vastness can be a literal experience of something like the Grand Canyon or it can refer to conceptual vastness, such as the notion of eternity.

Essentially, any stimulus that exceeds a person's normal range of experience in one attribute or another could lead to the perception of vastness, as can stimuli that lead people to feel as if they are part of something larger than themselves (Allen, 2018, p. 8).

Through scope, size, or complexity and across elicitors of awe, vastness expands and challenges people's frames of reference and ultimately serves a common denominator for this emotional experience.

Encountering Awe:

Despite its benefits, getting people to encounter awe remains fairly elusive. One study highlights this challenge by exploring the role of awe and solitude in urban outdoor settings. Relying on a range of research, which a) links awe to pro-environmental sentiments, b) suggests a short walk "can reduce feelings of rumination with awe acting as a mediator" (Mateer, 2022, p. 6), and c) emphasizes the positive cognitive impacts of awe-inspiring images of nature, this study looks at the ways in which awe and solitude can better connect people with the natural world. Through their work, Mateer (2022) suggests that

by challenging individuals' frames of reference (Bai et al., 2017), awe inspired by the natural environment may encourage individuals to contemplate the broader workings of the world and ways they fit into these systems (Pieper, 1963). Additionally, solitude in the outdoors may allow individuals to escape from the frenetic nature of their daily lives and provide them space for contemplation, an important aspect of leisure (Pieper, 1963) (pp. 4-5).

While outdoor leisure may serve as one entry point, awe walks offer an additional tool for encountering this emotional experience. An awe walk, which can be as brief as a 15-minute journey through an outdoor or indoor setting, is an opportunity to experience awe by finding novelty, appreciating the familiar, and embracing a childlike sense of wonder. In a study using these brief, intentional, and timebound explorations as an intervention, it was shown that participants who took awe walks, as compared to those who took regular walks, experienced greater awe and an increase in positive emotions. Moreover, participants who took more awe walks compared to other awe-walkers in the study experienced greater gains in awe while on their walks as well as larger increases in positive emotions.

2.2 Rhythmanalysis

Time, Place and Rhythms:

Written in 2004, Henri Lefebvre's *Rhythmanalysis: Space, Time and Everyday Life* explores the interrelation of time and space, introducing the notion of "localised time" or "temporalized place" (Lefebvre, 2004, p. 89). Building on this combination of space and time, according to Lefebvre (2004), rhythms are defined as the "interaction between a place, a time and an expenditure of energy" (p. 15). The subsequent investigation of these rhythms suggests that "places are always in a process of becoming, seething with emergent properties (Edensor, 2010, p. 3). Through rhythmanalysis the present, past, and future are intertwined. People, places and things avoid reification and are reanimated "by the dynamic temporality of the world and the processual character of place" (Edensor, 2010, p. 18). Although rhythms are viewed as fundamentally dynamic entities, many, through their relative slowness in comparison to the human lifespan and "regular patterns of flow," offer a sense of stability, a "consistency to place and landscape over time" (Edensor, 2010, p. 3), and ultimately "provide a backdrop to life against which the usual and unusual can unfold" (Edensor, 2010, p. 14).

The Rhythmanalyst:

Attuning to the "specific measure: speed, frequency, consistency" (Lefebvre, 2004, p.10) of each rhythm fundamentally involves a connection to the body. As Lefebvre explains, the rhythmanalyst must first listen to their body and learn rhythms from it in order to appreciate external ones (2004, p. 19). They are always "listening out," and using their body as metronome in order to be "capable of listening to a house, a street, a town as one listens to a symphony, an opera." (Lefebvre, 2004, p. 87). Put plainly, "a disembodied appreciation of rhythms is impossible" (Edensor, 2010, p. 5). According to scholars, the rhythmanalyst's prioritization of embodiment and rhythms of the self (those which are immediate, spontaneous, and inward) over rhythms of the other (those which are formal and public) is regarded as a form of resistance grounded in authenticity, conviviality, and variety. For the rhythmanalyst, the work of reclaiming rhythms is aimed at enhancing aesthetic and sensory experience and ensuring that "spaces of slowness supplement fast spaces in public and private domains." Furthermore, it requires a "a commitment to occupy time more attentively" and a "negotiation with everyday rhythms" (Edensor, 2010, p. 17).

Everyday Life:

Despite the fact Lefebvre emphasizes the ways in which everyday life is often rendered banal and meaningless through the rhythms of capitalism (Edensor, 2010, p. 13), it is nonetheless an important component of rhythmanalysis for it is "a site for the unfolding of multiple rhythms" and home to the tension between habit and disruption, stable and dynamic, repetition and difference. Within the

Conceptual Foundations for Awe

workings of our towns and cities [and] in the collision of natural biological and social timescales, the rhythms of our bodies and society, the analysis of rhythms provides a privileged insight into the question of everyday life (Elden, 2004, p. viii).

In order to begin understanding the ways in which rhythmanalysis can be applied to everyday life, Lefebvre provides a few essential concepts to ground his theory, including the notion of linear and cyclical time, as well as a classification of polyrhythmic, eurythmic, and arrhythmic rhythms. Through the everyday, linear time most noticeably appears as clock time while cyclical time appears as lived time (Elden, 2004, p. x). Linear rhythms are distinct from cyclical rhythms in that they originate from human and social activities and define themselves through the “consecution and reproduction of the same phenomena” (Lefebvre, 2004, p. 90). Meanwhile, cyclical rhythm “originates in the cosmic, in nature: days, nights, seasons, the waves and tides of the sea, monthly cycles, etc” (Lefebvre, 2004, p. 8). and unlike linear rhythms do not oppose the process of change or becoming. The relationship between these two forms of time is one of perpetual interaction and “antagonistic unity” whereby there is “sometimes compromise, sometimes disruption” (Lefebvre, 2004, p. 76). In the context of everyday life, Lefebvre’s classification of rhythms into polyrhythmic, eurythmic, and arrhythmic helps to address the multiplicity of rhythms that exist. Polyrhythmia is a state composed of diverse rhythms and “thus the everyday reveals itself to be a polyrhythmia from the first listening” (Lefebvre, 2004, p. 16). Eurhythmia is considered a state of alignment or equilibrium of rhythms, where multiple rhythms fit together harmoniously, whereas arrhythmia is characterized by discordance, disruption, and conflict and is a state in which rhythms “break apart, alter and bypass synchronization” (Lefebvre, 2004, p. 67).

Rhythm & Cities:

Lefebvre explains that “in order to grasp the rhythms, a bit of time, a sort of meditation on time, the city, people, is required” (2004 , p. 30). Cities offer a ideal site of exploration for rhythmanalysis given that public life is organized around all kind of exchange and interaction, material and non-material entities, signs and products. In other words; the city is organized by and around rhythms. Cities are a site of ongoing domestic, commercial, ecological, geomorphologic, political, social, and cultural rhythms that ebb and flow between polyrhythmic, eurythmic, and arrhythmic assemblages, all of which leave their mark on the everyday. Within the city, shops, bars, cafes, garages and so forth are meeting points at which individual paths congregate, providing geographies of communality and continuity within which social activities are co-coordinated and synchronized. The ongoing mapping of space through repetitive, collective choreographies of congregation, interaction, rest and relaxation produce situated rhythms through which time and space are stitched together (Edensor, 2010, p. 8).

This stitching together of rhythms through everyday activities and encounters form a “mutually constitutive” relationship between cities and bodies. As is true for the rhythmanalyst, however, this relationship is mediated through our sensory experiences with the world. The polyrhythmic dance of cities, “all amalgamate

through our embodied perception to create a sense of place” also known by Lefebvre as the ‘lived space.’

Although, this subjective and experiential aspect of city life is “a feature often forgotten in the ocularcentric Western society”, the lens of rhythmanalysis provides an entry point by which to “rehabilitate our sensory perception” and embrace the city as “an ephemeral, continually changing and fluid space of experience” (Degen, 2010, p. 23). Additionally, the application of rhythmanalysis to the built environment can provide a tool to better understand how certain features, like alleyways or staircases, act not only as a link between spaces but also a link between times. These connective elements of the city landscape, for example, provide a bridge between “time of architecture (the house, the enclosure) and urban time (the street, the open space, the square and the monuments)” (Lefebvre, 2004, p. 97), and highlight the experience of passing from one rhythm to another.

Rhythm & Awe:

Emphasizing the ways in which all forms and experiences are composed of rhythms, Lefebvre anecdotally shares a stream-of-consciousness description in *Rhythmanalysis*:

(This garden that I have before my eyes appears differently to me now from a moment ago. I have understood the rhythms: trees, flowers, birds and insects. They form a polyrhythmia with the surroundings: the simultaneity of the present (therefore of presence), the apparent immobility that contains one thousand and one movements, etc. . . .) (Lefebvre, 2004, p. 17).

The value of applying the concept of rhythmanalysis to the study of awe is that it allows experiences of awe—which so often involve trees, flowers, birds, and insects– to be perceived as dynamic, often repeating, rhythms rather than static, singular events. Rhythmanalysis, much like awe, helps to resituate and reconceptualize the self but does so amidst a range of polyrhythmic, eurythmic, and arrhythmic encounters that vary based on size, scale, and complexity. Like awe, rhythm is an embodied experience that animates reality in such a way that “the non-human dimensions of place [can no longer be considered]...a passive backdrop upon which human activity unfolds” (Edensor, 2010, p. 8). Recognition of the polyrhythmic assemblages of cities and space, similar to awe, creates an opportunity and need for accommodation. Furthermore, applying rhythmanalysis to awe not only offers a lens through which view the everyday but also the vastness of climatic, geomorphological and even cultural rhythms which seldom map on to the temporality of the human lifespan. In this way, rhythmanalysis is a critical framework for understanding awe not as an isolated emotional response, but as an emergent, embodied attunement to the shifting rhythms that constitute urban life.

2.3 Sacredness and Awe

Connections to Spiritual and Religion:

Awe is a spiritual emotion rooted in the notion of transcendence and connection, and has been “theorised by many to facilitate spiritual transformation and religious conversion (Jiang et al, 2018, p.2). Facilitation of this transformation is likely mediated by awe’s ability to help people 1) see themselves as part of a larger system, 2) recognize and appreciate things beyond the self, 3) reassess goals and provide new perspective on one’s meaning, 4) consider deeper questions and open their minds to what the universe has to offer, and 5) accommodate new information and points of view. Specific triggers of awe as they relate to spiritual and religion include perceiving the sacred in nature, unique moments of silence and understanding, crises, the miracle of life and death, encounters with outstanding or inspiring individuals, sacred buildings, beauty, and spiritual practices/ceremonies (Büssing, 2021, p. 6). Awe’s ability to increase spiritual and religious feelings has been identified through a variety of studies. Ethnographic research by Curtin (2009) indicated that awe and wonderment sparks spiritual feelings in tourists. An experimental study by Van Cappellen & Saroglou (2011) suggests that inducing people to feel awe increases participants’ religious and spirituality-related feelings and behavioral intentions (p. 11). Work by Valdesolo & Graham (2014) indicates that awe—mediated by the ways in which it decreases tolerance for uncertainty—increases supernatural belief and the role of nonhuman agents (p. 177). Additionally, Saroglou et al. (2008) provides evidence that exposure to self-transcendent and awe related information may lead to an increase in religious and spiritual beliefs.

Sacred Experience and Place in Cities:

Given the intersection of awe and spirituality and religion, this study pulls from Karilyn Emilie Krasin’s 1972 Master in City Planning thesis titled, *Sacred Experience and Place in Cities*. Focusing on the possibility of sacred environments, Krasin’s work highlights the quality of sacredness and as well as the spatial and physical conditions necessary to produce these experiences. According to Krasin, sacred experiences are accessible to everyone, both direct and personal, and characterized by “a sharp break from daily routine” (p. 23). Sacred moments involve a “heightened sensory awareness and the sense of suspended time” (p.23), allow for “the bonding of one’s inner world with [their] outside world” (p. 9), and provoke a “feeling of connection with the web of life” (p. 23). These sacred experiences can come in the form of gentle, ephemeral moments, which “should be a part of our daily lives” or, alternatively, as intense, disruptive events, marked by joyous surprise (p. 23). Krasin speculates on the nature of the everyday sacred moments suggesting that they may come from “rustling trees, surfaces that echo rain sounds, wind tunnels...or special chime for sunrise and sunset, or for a harbor city, for high tide and low tide” (p. 24). Meanwhile, Krasin suggest that the more momentous sacred occasions may take the form of a “celebration of a certain place and moment...enacted as a street dance, a formal pageant to lead persons to a given site or event; a feast of local

foods and wines; or a place reserved for solitary meditation” (p. 24). The result of experiencing such moments of sacredness include awareness of the line between immersion into chaos or profound meaning, and the need to give up, though perhaps only temporarily, on our old patterns and order (p. 21). Expounding on this need to let go, Krasin argues that through the sacred experience one must “abandon [their] fixed world momentarily; attach new meaning through direct experience to an object or place; [and] reorder [their] perceptions according to the insights provided by this new framework” (p. 66). In contrast with the typical urban experience—one that separates us from sources of production and experience, that “interrupts or short-cuts the complete organic cycle” (p. 67), and which tends towards “a complete and rational order to its parts” (p. 67)—the sacred experience “offers the possibility for disassembling the ‘packaged’ attributes of our urban life” (p. 67) so that we can examine, reorder and reinsert the fragmented parts. Furthermore, sacred experiences provide a glimpse into the ‘sacred present.’ Occurring upon the backdrop of “the harmony and mystery of all living things” (p. 86), the sacred present reflects a state of being caught up in, and filled with, the web of life. Krasin suggests that the sacred present can take place through celebration, meditation, and gardening and that it involves “the exploration of organic relationships where they exist, and the allowance for letting other relationships change and create their own order.” (p. 86).

As discussed in Krasin’s thesis, the experience of sacredness is supported and made possible by particular spatial and qualitative features. The sacredness of a place can be emphasized through a strong sense of spatial separation or boundary that helps to either separate the sacred from the profane or concentrate the sacredness within. Emphasizing this point, Krasin offers that,

the sacred potential might be more likely found within a particular site, where the change is abrupt, as in a cluster of closely-spaced buildings in a generally spacious neighborhood; or where the change is gentle, as in the hill crevice in a park. The feeling of spatial change may be supported by sound qualities, such as a change from clatter to stillness (p. 55).

Spatial transitions and boundaries, which can be marked by a clustering of buildings, walking paths, walls, or an arrangement of trees in a park, help to support a sacred experience by distinguishing a place from its utilitarian functions and denoting a clear distinction between its qualitative attributes (i.e. the residential and commercial, the unfamiliar and familiar, the playful and the serious). Within these boundaries can exist sacred clearings, where an abundance of life is present, as well as sanctuaries, or places where one goes when “our understanding of shared cosmos necessitates solitude, reflection, and private time to explore our personal part in the balance of life” (p. 87). Thresholds and doorways can reinforce spatial separation by symbolically or literally acting as transition points, or passageway from one state or identity to another. In addition to spatial boundaries, concentricity, on a site scale or city scale, can help to facilitate sacredness by symbolically or incrementally drawing attention to a focal point. Relying on prominent natural features of the land, geomorphic forms also support “the spiritual quality of ‘daily’ life by giving constant reminder of natural landscape rhythms” (p. 38). Helping to make sense of this array of sacred spatial attributes within the city, Krasin recommends incorporating roundness and stillness as general considerations, and more specifically “oddities with indeterminate functions, such as the small wood building flanked by skyscrapers, or the place that seems wild and overgrown in a trim setting.” Additionally, as a starting point for finding

Conceptual Foundations for Awe

sacredness, Krasin suggests looking for “dominant geographic characteristics, such as a harbor, or startling vistas” (p. 56) as well as more mundane places like abandoned sites, “sequential corridors, parks that link districts; productive or intimate gardens; and roofs and cellars” (p. 67).

Beyond their more physical attributes, a handful of qualitative elements can support the creation of sacred space. According to Krasin, one key element of sacredness is mystery. Found in places of “shadow and ambiguity, [and] in narrow passages” (p. 59), mystery allows for a sacred space to escape legibility and hold onto its secrets. This quality of sacredness can be emphasized through the juxtaposition of light and shadow, which may symbolically indicate a place’s movement from chaos or its return to elusiveness. Another qualitative aspect of sacred places is their relationship to permanence and renewal. Krasin advises that these sites should be “recreated periodically or buried” (p. 75), and describes that this repetition and renewal not only “suggests the powerful force of regeneration” (p. 58) but also “reflects certain cosmic rhythms that demonstrate this idea: the sun comes and goes, as do the tides, and the seasons” (p. 58). As a form of sacred maintenance, this effort of renewal and invocation of impermanence ensures a connection with these cosmic rhythms. In addition to mystery and renewal, the sacredness of space can be supported through its relationship to history. According to Krasin, “the sacred function of history is to get at the deep sources for our understanding, [and] to regenerate events that have meaning to us” (p. 73). This form of history is one not simply for remembering and instead is one that is lived. This sacred history can be participated in through traditions and rituals or can be brought back to life through rediscovery, reimagination, or patterns and cycles that have been forgotten.

In summary, Krasin provides recommendations for how designers and planners can and should integrate sacred experience into cities. This includes 1) offering direct and personal experience that can provide a basis for “examining stereotypes of meaning” (p. 73), 2) promoting “a feeling for the organic interdependence of the city as a human environment” (p. 73), 3) working to integrate our subjective and objective worlds, 4) providing a context for both for mystery and magic, 5) balancing free spaces at various scales that invite a range of life with spaces and activities designed to direct attention to “the constant, cosmic elements that demonstrate regeneration of life” (p. 87), 6) pulling us “towards one another in a spirit of celebration [and] in a harmony with the land in which we live” (p. 80), and 7) finding a ‘simple center’ by weeding out noise and distraction before deciding to elaborate. From a more cautionary perspective, Krasin suggests that designers and planners should 1) “learn to recognize the profane, and leave it alone; or use it as the ‘chaos’ out of which the sacred can take form” (p. 76), 2) work “to release, direct or suggest “the potentialities of sacredness, not to ‘capture’ them” (p. 76), 3) acknowledge the potential ramifications of trifling with the sacred magic and take responsibility “for leaving some possibilities alone” (p. 76), and 4) “not to insist on very remote traditions, or completely new ones, but to let the appropriate ones grow in their own time” (p. 76).

Excerpt from Sacred Experience and Place in Cities:

“As I imagine it, the designer-shaman should have comets and half-moons dripping from the folds of his or her sleeves, and a tool kit that includes: honey and linden, the Titanium Platform of the 7th Starbowl, the clear cool green of a spring morning, fireflies in the shadows of alleys, velvet moss and dream herb, dim moon-eyed fishes, the light of the sun through silk, imaginary gardens with real toads in them, and the oldest rock in the deepest strata of the Earth’s mantle. He should be able to close off the abrasive clatter of impatient cars, the piles of rat-chewed litter collecting in the ruts of oily alleys, the hum of the neon sign, and the intrusive megaphone blaring a worthy cause. Instead, he should catch at the peculiar skinny shadow that alleys often have, and their ‘sucking’ labyrinthine quality; the sun dancing across a row of office windows at sunset, filling them each in turn with a moment of its own image; the sea smell that collects in a doorway far from the harbor. He should poke around in the weedy lot, the crumbling ruin, for evidence of ‘aliveness’ and for a clue as to how things arrange themselves when left on their own. Then, in abstracting his search, he should look at these aspects of order in the city framework: the places things begin, the transition points, and the places that border activities. He should look for a certain disorder as well, as evidence of transitions. He should also seek out things that are immune to rational coding, that have emotion, poetry, magic, and vitality: the spirit of the counterculture; the fantasy of Gaudi’s monuments; the song of the wind, the tracks of certain stars. He might continue by searching out the curves and corners, the pockets and bumps he has discarded from his own previous designs, and reconsider them as ‘commas’ in a sacred vocabulary, or ‘small cracks’ between the worlds that fit into no rational scheme. Then, he should also look at the things at the root of the big ‘rational elements’--the skyscraper foundation, the sea bottom, the heart of the furnace. The function of such a search is to loosen the designer’s own imagination, to find the poetry in things which he has previously considered primarily in terms of utility” (pp. 78-79).

Section Three



Methodology

“Awe seems to resist precise definition and measurement, the bedrock of science...How could scientists lead people to feel it on cue and measure its near-ineffable qualities, or document how awe transforms our lives, if indeed it does?

- Dacher Keltner, 2023, *Awe*, p. 6



3.1 Research Design and Data Collection

Without any research precedents or proxy variables to rely on which might serve as a baseline for assessing awe specifically within urban environments, this project drew upon the qualitative research conducted by Dacher Keltner. Relying on 2,600 narratives across 26 languages, Keltner’s (2023) extensive investigation of awe revealed a taxonomy through which to classify awe known as the ‘eight wonders of life.’ The discovery of these ‘wonders’ indicate that across cultures, awe is consistently, though not exclusively, found in the following forms: moral beauty, collective effervescence, nature, music, visual design, spirituality/religion, life & death, and epiphany. Given the scope and consistency of these findings, it was reasonable to assume that these forms of awe could inform an approach to understanding the intersection of awe and public space in an urban context.

In efforts to prioritize finding awe within the the time allotted for this project, Cambridge, Massachusetts was chosen as the primary unit of spatial analysis and the ‘eight wonders of life’ were used to determine research participants. The underlying assumption of focusing on people associated with these eight forms of awe was that they might be more inclined to experience awe and might have more to say about the role of awe and public spaces. To test this assumption, however, a second category of potential participants was chosen based on their lack of association with these ‘wonders of life.’ Various participant groups were identified based on existing knowledge of the area and internet searches. Potential awe-related groups based on the ‘eight wonders of life’ included: moral beauty (meditation centers, advocacy groups, bike safety organizations), collective effervescence (YMCA, community art center, dance groups), nature (community gardeners, conservation groups), music (community chorus, jam groups), visual design (museum docents, art association), spirituality/religion (churches, synagogues), life & death (Death Cafe, birth centers, hospice care), and epiphanies (adult education centers). Potential non-awe related groups included: redevelopment authorities, neighborhood associations, library groups, and city commissions.

Outreach to all potential participants consisted of a cold call or email with an invitation to partake in a research study that was exploring how people experience moments of awe in Cambridge. All participants were asked to partake in a data collection process that involved a 30-minute initial interview, a one-month awe intervention period that included an awe walk, and a follow-up 30-min interview. This method of documenting awe was a combination between Kelner’s narrative-based research and Gehl’s experiential investigation of public space and awe in New York City and Charlotte, NC which involved personal observation, interviews, and participatory mapping. The objective behind an awe intervention and follow-up interview was to see if introducing a direct experience of awe impacted awe-related and non awe-related groups differently. The awe walk was an obvious intervention for three reasons, 1) as discussed in *Awe* by Dacher Keltner, “walking engages an awe-like form of consciousness,” 2) awe walks are an increasingly well known practice with documented benefits such as more positive emotions and less daily distress (Friedlander, 2026), and 3) awe walks are an accessible intervention within an urban environment.

Ultimately, six participants associated with the nature ‘wonder of life’ and five participants unrelated to a ‘wonder of life’ signed up for the study. Each participant of the 11 participants took part in a

Research at a Glance:

<u>Data Collected:</u>	<u>Technical Analysis:</u>
22 semi-structured interviews 10 awe walks 220 locations of awe - Personal awe data	- Qualitative coding and application of conceptual framework - Frequency and pairing analysis - Multiple Correspondence Analysis & Hierarchical Clustering - Spatial autocorrelation - Group comparisons

semi-structured initial and follow-up interview. The initial interview protocol consisted of 10 questions about subjects’ relationship to Cambridge, their use of public spaces, as well as their past experiences of awe and the spatial quality of these experiences. The follow-up interview consisted of 10 questions regarding subjects’ experiences of awe since the first interview, their experience of going on an awe walk, and whether their experiences in the study would have a lasting impact on their relationship to Cambridge (refer to Appendix A for a full list of the initial and follow-up interview questions).

To supplement findings from both research groups, personal observations of past experiences of awe in Cambridge were collected and documented. Additionally, as means of better understanding the experience of awe and and to gain familiarity with the research methodology, personal awe walks were conducted every two weeks over the span of two months in neighborhoods throughout Cambridge.

3.2 Analytical Methods

The process of creating a framework for awe was grounded in the experience of participants in this study. During the interview process, participants were asked to share what moments of awe they had experienced and where exactly these experiences took place. To direct the conversation, they were prompted to think about these moments of awe based on four characteristics: interconnection, vastness, wonder, and mystery. These categories were informed by Dacher Keltner’s *Awe* and were used as a way to get participants thinking about awe in a way that didn’t limit their conceptions of this experience or simply narrow it down to one of the ‘eight wonders of life.’ After each interview, instances of awe were recorded as spatial points in Google Maps. Each point included a name, relevant pictures, notes about this experience of awe, as well as which awe-related characteristics these points corresponded to. Data points were color coded based on participant group as well as whether each instance of awe came from the initial or follow-up interview. Often times, recorded instances of awe invoked multiple dimensions yet for the sake of simplicity only a primary elicitor was chosen. As was hoped for, participant shared experiences of awe that took place across a wide range of locations and that were marked by a variety of descriptors extending beyond the four characteristics of awe involved in the interview protocol.

Accounting for these added dimensions of awe involved exporting point data from Google Maps and analyzing this information in the form of a CSV. Latitude and longitude data as well as names for these experiences of awe were preserved while notes and descriptions were subject to interpretation and categorization. In reading through the over 200 data points and their respective descriptions, key details and questions began to surface about the nature of awe, including:

- 1) Geographies of awe
 - ❖ Where exactly is awe being experienced in the built environment? What type of movement or transportation is associated with experience of awe?
- 2) Morphologies of awe
 - ❖ What are the external forms or stimuli (i.e. the people, places and things) that give to rise to awe?
- 3) Phenomenologies of awe
 - ❖ What are the perceptions and defining qualities of the experience of awe?
- 4) Temporalities of awe
 - ❖ What is the temporal nature of this experience of awe? Is it a singular event or is it part of a cycle? What is its potential for repetition?

Resolving these questions involved parsing through the descriptions to distinguish between the what, where and how of awe, creating and refining themes, and grounding the qualitative findings within the aforementioned conceptual frameworks. The following subsections provide methodological details about how these aspects of awe were categorized.

Geographies of Awe

Beginning to grapple with an analytical framework first involved determining the types of locations associated with each instance of awe. Given that all moments of awe were represented with XY-coordinates, this process simply involved identifying where exactly these moments of awe occurred and creating a consistent and efficient set of categories to capture meaningful differences in location types. In total, nine location types were chosen: civic; community and cultural venues; corridors; educational institutions; industry and office buildings; parks; plazas; residences; streets; and waterfronts.

Civic, community, and cultural venues included libraries, concert halls, cafes, markets, community centers, museums, cultural institutions, public lobbies, and places of worship. Corridors included train tracks and transportation right-of-ways. Educational institutions included universities and schools. Industry and office buildings included research labs, factories, and offices. Parks included open spaces, cemeteries, dog parks, community gardens, pocket parks, commons, and courtyards. Plazas included open urban spaces and public squares designed for gathering. Residences included personal dwellings. Streets included sidewalks, bike lanes, and roads. Waterfronts included beaches, canals, bridges, and linear paths along bodies of water.

In addition to these nine location types, each instance of awe was coded based on the transportation modality through which the awe occurred. These modes of transportation included: walking, biking, and public transit. This list of transportation modes is not exhaustive but reflective of those present in the data.



Morphologies of Awe

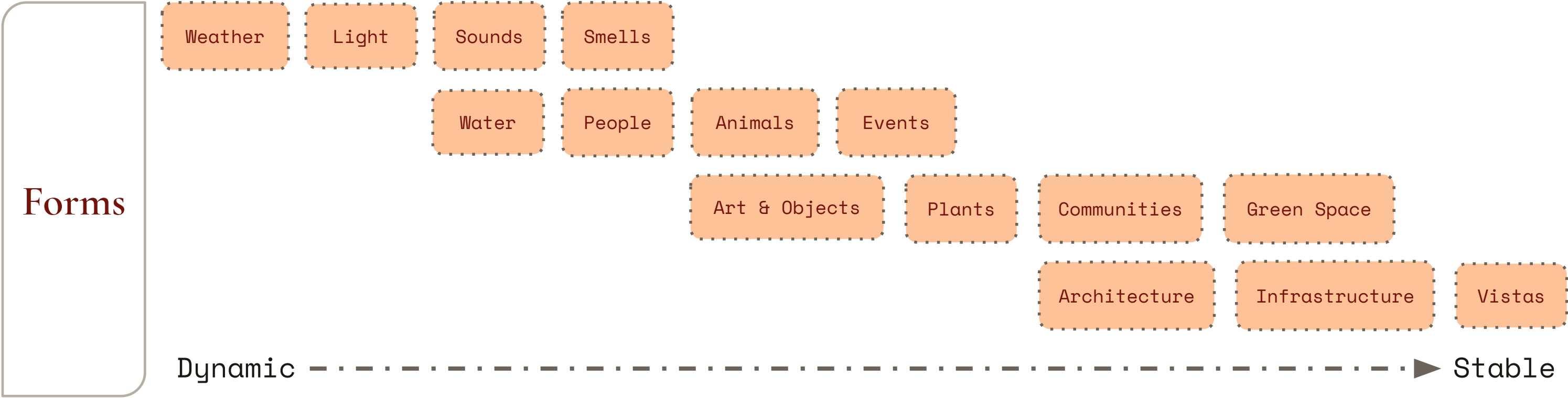
As became clear from the data and literature, experiences of awe in the built environment rely on external stimuli of many shapes and sizes. Moments of awe were found in the beauty of an old oak tree in a nearby park, the wonder of afternoon light at the library, or the mystery of a clandestine courtyard. Determining what morphologies provided the initial trigger for awe involved a handful of coding iterations to ensure that each moment of awe was properly identified and that forms of were efficiently categorized. This process revealed 15 distinct forms of awe: animals, architecture, art & objects, communities, events, green spaces, infrastructure, light, people, plants, smells, sounds, vistas, water, and weather. As a further classification of these forms, one which highlights the rhythmic quality of morphologies, they are presented on a spectrum of stable to dynamic.

The most dynamic of these forms included weather, light, sounds, smells. These are forms that are often subject to constant change, experienced through the senses, and which flow naturally through an environment. Weather included atmospheric and meteorological conditions like rain, snow, ice, frost, and seasonal transitions. Light included sunsets, reflections, shade, and seasonal lighting changes. Sound included auditory encounters created through movement or nature. Smells included olfactory encounters with food and nature. Less dynamic forms of awe were water, people, animals, and events. These were aspects of the environment that move more slowly and which rely on systems and relationships. Water included rivers, canals and ponds. People included strangers, individuals, employees, historic figures, and the presence of others through temporary gatherings. Animals included birds and mammals. Events included gatherings, performances, festivals, lectures, sporting events, celebrations, and public activities.

Increasingly stable forms of awe were art & objects, plants, communities, and green space. Though subject to change, these objects can take on a permanent quality through maintenance, care, and intentional practice. Art & objects included public art like murals and sculptures, installations, found items, and museum exhibits. Plants included trees, lawns, flowers, shrubs, gardens, and mushrooms. Communities included social bodies, organizations, and collections of people who have a shared purpose. Green space included designed or semi-designed outdoor spaces.

The most stable forms of awe included architecture, infrastructure, and vista. These three features of the environment are the least subject to change and take the longest to create. Architecture and infrastructure are human endeavors that require time and energy for planning and implementation, while vistas often rely on geomorphological processes. Architecture included building forms, enclosure, materiality, and ornamentation. Infrastructure included bridges, alleyways, boathouses, and transit spaces. Vistas included expansive views of often of the sky, cityscapes, and bodies of water.

Relying on the conceptual frameworks underlying this research, one additional morphological category was used to supplement the 15 identified forms of awe. This category was the ‘eight wonders of life,’ which include moral beauty, collective effervescence, nature, music, visual design, spirituality/religion, life & death, and epiphany. Using these classifications as potential morphologies helped to align the study with the science of awe and to see how these key elements of awe played out in Cambridge (refer to Conceptual Foundations for Awe section for a definition of the ‘eight wonders of life’).



Phenomenologies of Awe

The experience of awe is first and foremost an emotional, lived event. While it relies on some internal or external stimulus to invoke this feeling, it fundamentally a perception about the world that relies on attunement to the quality and changes of one’s inner state. An initial attempt to get at the quality of the internal experience of participants came through the characteristics of awe addressed in the interview protocol: interconnection, vastness, wonder, and mystery. Interconnection reflected the ways in which awe connects people to systems and helps them to feel like part of larger whole; vastness is core element awe and a recognition of something exceeding one’s ordinary level of experience; wonder is an emotion closely related to awe that involves curiosity and openness; and mystery is tied to the ways in which awe escapes the known and understood. During the interview process, these attributes served as prompts to draw out additional phenomenological qualities of awe and helped to reveal new features of awe like beauty and surprise, as well as notions like wildness, familiarity, and inclusivity. In order to preserve the information captured from each interview, the four characteristics of awe were coded as phenomenological categories and applied to each instance of awe. Due to how frequently it showed up across interviews, ‘beauty’ was coded as a fifth characteristic, signifying the reliability of aesthetic harmony in moments of awe.

As the study progressed and more instances of awe were documented, related qualities of awe existing across various spectrums began to surface. Informed by the tensions present in Rhythmanalysis such as cyclical vs. linear time, and eurythmia vs. arrhythmia, obvious pairings of awe were slowly established, refined based on similarities or differences, and kept so long as there was at least one instance of awe present for both sides of the pairing. Supported by the understanding that “awe arises via appraisals of stimuli that are vast, that transcend current frames of reference, and that require new schemata to accommodate what is being perceived” (Piff et al., 2015, p. 884), chosen pairings reflected potential vastness and subsequent accommodation across six registers of awe apparent in the data: formal, social, ecological, evaluative, temporal, and spatial. Respectively, these registers reflect six tensions: ornate vs. functional, inclusive vs. exclusive, wild vs. curated, impressive vs. ordinary, novel vs. familiar, and expansive vs. intimate. While multiple tensions could be present for a single instance of awe, this was uncommon. As such, as well as the desire to get at the phenomenological root of experiences of awe, each instance of awe was coded based a single tension.

The tension of ornate vs. functional focused on the aesthetic qualities of awe and often pointed to a expression of generous intentionality. It addresseed experiences of awe around things made unnecessarily beautiful, and those that achieve their purpose with simplicity and ease. The tension of inclusive vs. exclusive indicated the social qualities of awe and often denoted a sense of belonging or intrigue. It pertained to experiences of awe related to public and private boundaries, inclusion, accessibility, secrecy, and social joy. The tension of curated vs. wild emphasized the ecological qualities of awe and spoke to level of care, intentionality, and ecological agency in the built environment. The tension of impressive vs. ordinary highlighted evaluative qualities of awe and centered on experiences of people, places, and things that exceeded a standard or norm for their category. This tension also addressed the presence of everyday

experiences that are seldom categorized or compared, and which therefore provide an experience of pleasant appreciation. Lastly, the tension of novel vs. familiar engaged with the temporal quality of awe and was often predicated on an experiential baseline from which changes and transitions could be observed. It addressed genuinely new experiences and emphasized one’s relationship to a thing or place over time.

In addition to the characteristics and tensions of awe, four phenomenological aspects of awe were coded across the dataset: alterity, surprise, threshold, and sanctuary. Alterity and surprise were aspects of awe clearly expressed by participants whereas thresholds and sanctuaries were less explicit in the data. Both alterity and surprise were dimensions initially coded as tensions yet due to their relative scarcity compared to the five characteristics of awe and due the fact no opposites, such as sameness or predictability, were identified as sources of awe, they were eventually adopted as a binary attributes (i.e. yes or no).

Characteristics

Interconnection:

- The sense being a part of of a larger whole or seeing oneself as embedded in a system or network.
“Awe shifts us to a systems view of life” (Keltner, 2023, p. 248).

Vastness:

- The perception of something physically, temporally, conceptually, or social large that transcends or exceeds one’s current understanding.
“Awe is the feeling of being in the presence of something vast” (p. 7).

Wonder:

- The state of curiosity, openness, and receptivity.
“Tap into your childlike sense of wonder... approach what you see with fresh eyes, imagining that you are seeing it for the first time” (p. 105).

Mystery:

- The encounter with something that cannot easily be explained or understood.
“[Awe is a feeling that] requires the unexpected, moves us toward mystery and the unknown” (p. xvi).

Beauty:

- The perception of aesthetic harmony or sensory pleasure in the natural or built environment.
“In cities judged from photos to be more evocative of visual awe, people report more robust health” (p. 284).

Thresholds and sanctuaries were coded at the end of the analytical process as a way to describe particular experiences of awe and to further ground the study in the the conceptual framework of sacredness.

Alterity was a core phenomenological feature for certain instances of awe in Cambridge, appearing most often in encounters with plants and animals. Defined as an experience of irreducible otherness or an encounter with that which is beyond the self (Clegg & Bailey, 2008), alterity is a phenomenological attribute aligned with idea that “awe encourages other-oriented, prosocial behavior” (Piff et al., 2015, p. 896), related to a state of openness and receptivity, and involving recognition of animate interiority outside of oneself. Similarly, surprise, along with its cognitive correlate, unexpectedness, was a defining feature of many instances of awe. Considered a short-duration emotion that is a bridge between cognitive and emotions states, surprise is defined as a “feeling of mild astonishment or shock caused by something unexpected” (Brown, 2021), and as an analytical variable was often accompanied a change of routine, a sense of mystery or discovery, and moment of appreciation. Given the nature of it being a separate emotional response that occurs before or in parallel to awe, and due to the fact that it can serve as valence-like attribute to qualify the tone of each experience, surprise was coded as a binary phenomenology. The concept of thresholds and sanctuaries are derived from Krasin’s *Sacred Experience and Place in Cities* and operationalized in this

study in the form of binary phenomenological variables. Though these elements were not immediately obvious based on interviews with participants, they nonetheless helped to explain certain phenomenological aspects of awe and further grounded the study in notions of sacredness. Krasin likens a threshold to a doorway, a gate, or entry that acts is “symbolic of the to another state or identity or world” (1972, p. 53) and wistfully circles around the concept, explaining that a threshold might “might emerge as an horizon seemingly dropping into nowhere...[or] it might be an unseen force of wind that the person must turn slightly to pass through” (p. 94). For the purpose this study, the concept of threshold was applied positively to instances in which there is a sudden transition or fleeting quality associated with the experience of awe, or an appreciation of sensory experience associated with movement. The notion of sanctuary, while less clearly defined, is referenced throughout *Sacred Experience and Place in Cities*. Krasin suggests that sanctuaries are the most appropriate building form for when “our understanding of shared cosmos necessitates solitude, reflection, private time to explore our personal part in the balance of life” (p. 87). The concept of sanctuary was coded as a phenomenological binary for instances of awe associated with feelings of solitude, relaxation, coziness, or peacefulness, or in moments when the the source of awe related to intimate spaces, nooks and crannies, or enclosures that had sanctuary-like qualities.

Tensions

Ornate
↔ Functional

The tension between forms and spaces that exceed their function through decoration, craft, and aesthetic excess and those that perform their functions with elegance and fidelity. Aspects of this tension include: historical richness, material quality, and unnecessary beauty.

Heuristic: Does this thing do more than it needs to or exactly what it needs to?

Inclusive
↔ Exclusive

The tension between spaces and communities experienced as open and inviting and spaces experienced as inaccessible or closed, thereby generating awe through connection or and mystery. Aspects of this tension include: belonging, boundaries, intrigue, and social relations.

Heuristic: Does this space or group invite you in or hold you at a boundary?

Wild
↔ Curated

The tension between living things that exceed or escape human management, thereby asserting their own autonomy and liveliness, and those that are defined by human care, cultivation, and maintenance. Aspects of this tension include: intentionality, ecological agency, and stewardship.

Heuristic: Is this place managing nature or is nature managing itself?

Impressive
↔ Ordinary

The tension between things that exceed standards or expectation for their categories and those that pleasantly or simply stand outside of normal categorization. Aspects of this tension include: scale, ability, significance, and the everyday.

Heuristic: Does this thing exceed a normal range for its category or is it typically uncategorized?

Novel
↔ Familiar

The tension between encountering something genuinely new, or a known thing made new, and encountering something that is already known again. Aspects of this tension include: time, memory, attentiveness, and a relationship to place.

Heuristic: Is this something new being noticed, or is it being seen more deeply?

Expansive
↔ Intimate

The tension between spaces that highlight the body’s relationship towards something vast and outward and spaces that hold, enclose, or provide refuge. Aspects of this tension include: physical orientation, spatial scale, and containment.

Heuristic: Does this space open outwardly beyond you or does it help to draw you in?

Phenomenological Binaries

Alterity:

A binary category indicating that the experience of awe involves an encounter with irreducible otherness and a recognition of presence beyond oneself. Aspects include: receptivity and an animate other.

Heuristic: Does this thing help you to recognize and appreciate something beyond yourself?

Surprise:

A binary category indicating that the experience of awe is unanticipated or caused participants to redirect their attention. Aspects include: discovery and disruption.

Heuristic: Was this experience of awe unexpected, did it interrupt, or did it bring you into the present moment?

Threshold:

A binary category indicating that the experience of awe involves a felt sense of transition, or movement across spatial, sensory or perceptual register. Aspects include: transformation, passage, and liminal spaces or moments.

Heuristic: Does this place or experience denote a passage from one quality to another?

Sanctuary:

A binary category indicating that the experience of awe is accompanied by a felt sense or potential sense of enclosure and refuge. Aspects include: solitude, shelter, and intimacy.

Heuristic: Does this place feel like it's drawing you in rather than pulling you outward?

History:

A binary category indicating that the experience of awe is accompanied by a felt sense of the past or an awareness of the accumulation of time. Aspects include: longevity, connection to certain time periods, and the presence of time.

Heuristic: Is this moment of awe informed by the past or connected to something old?

Temporal Binary

Temporalities of Awe

The need to incorporate a temporal analysis was made evident when trying to decide how to code multiple instances of awe in the same location. A primary goal of this project was to understand spatial aspects of awe, yet analyzing solely on the basis of spatiality meant eliminating the temporal differences reflected across instances of awe. For example, Mount Auburn Cemetery was listed as a location of awe by two participants. One described a sense of awe due to the vastness evoked by the ecology, burial plots, and monuments. Another described a sense of wonder from the Solstice Winter Lights Festival that occurs each year and which transforms the cemetery through light and sound art installations. Addressing the inherent temporality of awe while still incorporating a spatial lens involved operationalizing the conceptual framework of Lefebvre's Rhythmanalysis. In *Rhythmanalysis*, Lefebvre explains that "everywhere where there is interaction between a place, a time and an expenditure of energy, there is rhythm" (2004, p. 15). Understanding awe as an interaction between place, time, and expenditure of energy—which in the case of awe is most commonly the act of noticing—not only provided a basis for addressing the multiplicity of awe in a single location but also a way to reimagine instances of awe across time.

The temporal and rhythmic quality of this emotional experience was ultimately confronted through its frequency, or potential for recurrence. Categories of awe temporalities were inductively identified and refined, and each instance of awe was coded based on whether the experience was ephemeral, periodic, seasonal, recurring, daily, or timeless. In attempt to build a bridge with the spatial forms of awe, these temporalities are also presented on a spectrum of dynamic to stable. Ephemeral awe reflected instances that were considered 'once in a lifetime' or that had a very small chance of happening again. Periodic awe indicated moments that had the potential to happen again but on an unpredictable basis. Recurring awe referred to instances happening on a weekly, month, yearly, or other repeating schedule and was intended to align with Lefebvre's notion of linear time, or time that is derived from social practice and human activity (2004, p. 8). Seasonal awe encompassed moments determined by natural rhythms and was intended to align with Lefebvre's notion of cyclical time, or time that "originates in the cosmos, in the worldly, in nature" (p. 76). Daily awe captured moments that were subject to diurnal rhythms, such as the shade created by movement of the sun, and those that were available everyday but within time-bounded conditions, like a museum's open hours. Lastly, the category of timeless awe was applied to instances connected to elements in the environment which were continuously available, like views of the city skyline, buildings, and murals.

Given how often it was mentioned, the notion of history was included as a temporality of awe. Rather than a frequency, history was considered a temporal binary and was applied positively to instances that referenced historical features or the importance of age. While history was inductively chosen as an aspect of awe, the inclusion of it as an analytical variable helps to further situate the study within *Sacred Experience and Place in Cities*. Krasin emphasizes "the need for a history we can use, not simply remember" (1972, p. 73) and suggests that "history, in the sacred sense, is one that is still alive, one in which we can participate" (p. 80). By acknowledging history as an active feature of awe, it allows for invocation of its sacredness and creates an opportunity for participation through celebration and remembrance.

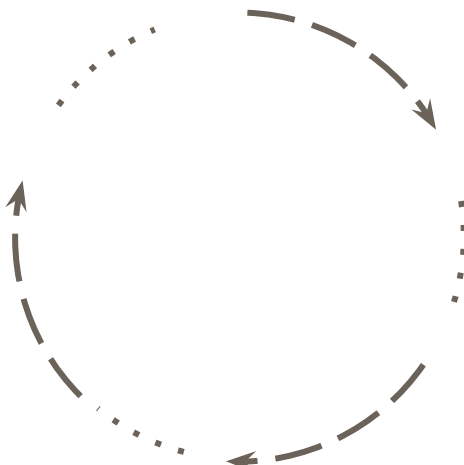
Frequencies



Ephemeral

A category indicating that the experience of awe occurred once and could not reliably be recreated. Aspects include: rarity, and singularity.

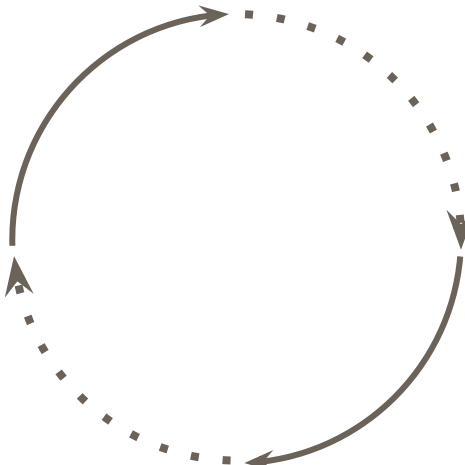
Heuristic: Is this moment of awe likely to occur just once?



Periodic

A category indicating that the experience of awe is part of an unpredictable or inconsistent cycle. Aspects include: irregularity and synchronicity.

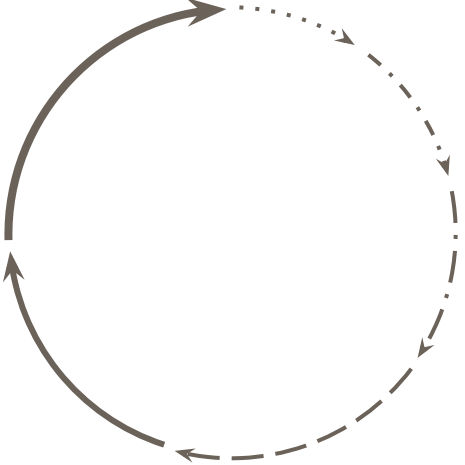
Heuristic: Does this moment of awe return on an unidentifiable or sporadic cycle?



Recurring

A category indicating that the experience of awe is available based on an human-created timeframe. Aspects include: programming and celebration.

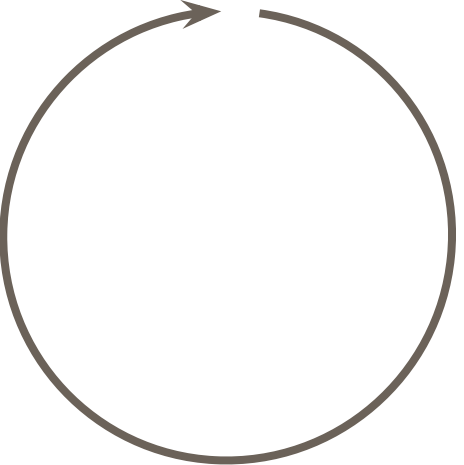
Heuristic: Does this moment of awe recur based on a calendar or schedule?



Seasonal

A category indicating that the experience of awe is dependent upon natural cycles. Aspects include: migration, fluorescence, and weather.

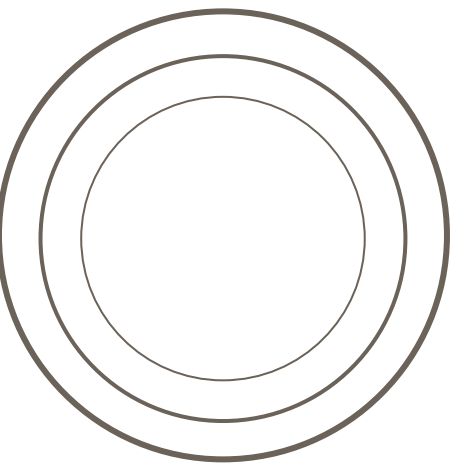
Heuristic: Does this moment of awe return with the seasons?



Daily

A category indicating that the experience of awe is marked by everyday rhythms. Aspects include: habit, commerce, work, and diurnality.

Heuristic: Is this moment of awe available within the rhythm of an ordinary day?



Timeless

A category indicating that the experience of awe is available continuously, regardless of season or schedule. Aspects include: persistence and durability.

Heuristic: Is this awe available in any given moment?

Dynamic —————> Stable

3.3 Quantitative Methods

In order to make sense of the 220 instances of awe categorized through the analytical framework, several quantitative methods were used. First, to determine which specific variables within each category (for example, a specific characteristic of awe like beauty and a specific temporal quality like seasonality) paired together more or less often than would be expected by chance, standardized Pearson residuals were calculated for each predictor–outcome pairing across all four analytical domains: geography, morphology, phenomenology, and temporality. Residual values exceeding $|z| > 2$ (approximately $p < 0.05$) were treated as statistically meaningful, thereby helping to identify which specific variables of awe consistently co-occurred and which ones rarely co-occurred. The strength of these pairings ($|z| > 2$) was assessed alongside their sample size, and pairings with $n \geq 5$ were included in the analysis. A matrix of all all z-scores and n-values broken down by tensions of awe is provided in Appendix B.

Next, to identify broader patterns across the categorical data, Multiple Correspondence Analysis (MCA) was used to translate the 220 awe experiences into a numeric structure that allowed similarity between experiences of awe to be measured. Hierarchical Clustering on Principal Components (HCPC) was then applied as a way to group the most similar experiences together, producing four distinct clusters of awe. These clusters, interpreted as archetype of awe, were characterized by a distinct combination of geographic, morphologic, phenomenological, and temporal variables. To assess where and to what degree each of these archetypes were spatially clustered within Cambridge, global and local Moran's I statistics (Moran, 1950) were calculated. These tests provided a basis to determine whether experiences of awe belonging to the same archetype tended to occur near one another in the city, or whether they were scattered even throughout it. From this analysis, a number of awe hotspots were located and that helped point to a set of anchoring institutions, parks, and neighborhoods. Finally, to compare how each of the participant groups and personal awe instances differed from one another, each group was examined against the four archetypes of awe to see whether some were more common for one group than another. From here, a chi-square test was used to determine whether or not these intra-group differences were statistically meaningful. Analyses were conducted in R 4.4 (R Core Team, 2024) using the FactoMineR (Lê et al., 2008) and spdep (Bivand et al., 2013) packages.

The quantitative approach to this study was developed with the support of an AI language model, Claude (Anthropic, 2026). Through an iterative process, appropriate statistical techniques were identified for categorical data, code with RStudio was generated and debugged, and findings were refined and discussed. All analytical decisions, code review, statistical results, and written interpretations are the author's own.

Section Four



Cantabrigian Awe

In moments of awe, then, we shift from the sense that we are solely in charge of our own fate and striving against others to feeling we are part of a community, sharing essential qualities, interdependent and collaborating.

- Dacher Keltner, *Awe*, 2023, p. 40

4.1 Qualitative Findings

Discovering the qualitative nature of Cantabrigian awe relied on thematic analysis of 22 interview and 10 awe walks across both participant groups. This involved identifying specific ideas and elicitors of awe from both interview phases and, when relevant, organizing these according to the registers of awe associated with phenomenological tensions: formal, social, ecological, evaluative, temporal, and spatial. It is worth noting that experiences of “awfulness” are also enumerated as they routinely came up in response to the follow-up interview question–“what places did you think to avoid on your awe walk?”– and unprompted during the initial interview phase. Supplementing this analysis of the interview process is a summary of participant’s experiences from the awe walk intervention, along with any final reflections regarding shifts in participants’ experiences of awe and of Cambridge. To protect participant confidentiality, quotations in this section are attributed generically and any identifying contextual details have been removed or generalized.

The qualitative analysis sheds light on the commonalities and difference across groups – the awe-related group more commonly referenced experiences of awe related to ecological and social factors, while non awe-related participants tended to cite more social, spatial, and temporal aspects of awe. Meanwhile, analysis of the awe walk intervention and final reflections reveal that participants experienced an increase sense of well-being, an increase connection to Cambridge, as well as a desire to seek out more experiences of awe.

Interview Insights:

Participants reported:

- Ecological encounters and seasonal changes
- A sense of connection, belonging and mystery through nature and social life
- Importance of social conditions like safety, inclusivity, and neighborliness
- Sensitivity to time and change through history, legacy, development and daily life
- Attention towards aesthetic and experiential qualities like materiality, whimsy, and discovery
- Appreciation for urban forms and qualities like squares, pocket parks, interstitial spaces, and walkability
- Experiences of ‘awfulness’ tied to traffic, noise, lack of safety, inaccessibility, and exclusion

Interviews

Awe-related Participant Group:

The awe-related group involved six participants who were chosen based on their connection to one of the key elicitors of awe, and wonder of life, nature. When asked about what informs their experience of awe during the initial interview, awe-related participants most often highlighted ecological and social qualities. From an ecological perspective, participants talked about:

- the resilience and mystery of plants
- the soothing and reassuring nature of observing plants
- the beauty of both manicured and wild places
- a sense of belonging gained through familiarity with plants and animals
- connection with nature as something that isn’t fully knowable
- a relationship with bodies of water like the Charles River and Fresh Pond
- the connection between the river and ocean
- vistas of the sky and clouds
- the moments of ‘immediacy’ and ‘profound pleasure’ of gardening
- a sense that ‘awe happens right at the level of the soil’
- Sunsets, reflections, and shade
- weather and the transition of the seasons

From a social standpoint participants mentioned:

- free and inclusive activities
- safety and care
- active street life

- a diversity of amenities that bring people together
- the importance of neighborliness and people’s willingness to contribute to the greater good
- the sharing of resources
- a willingness to create beauty and ensure that it is both visible and accessible
- pleasurable moments of ‘cross-cultural confusion’
- familiarity with shopkeepers and fellow customers

While less prominent for this group, their appreciation of the spatial and aesthetic qualities of awe revolved around:

- the connection of Cambridge’s squares
- walkability
- the value of cozy pocket parks and small spaces within larger parks
- light and color
- solitude
- learning new skills
- seamless transitions
- being able to contribute to the unfolding and creation of spaces
- public spaces and thruways that create more connectivity
- spaces that are not just beautiful but also useful

Qualities of Cambridge that invoked a sense of awfulness included:

- streets with lots of cars

Cantabrigian Awe

- loud noises and traffic
- self-centeredness
- debris and trash
- disconnection or interrupted connectivity in the built environment
- the presence of spaces that don't accommodate people or that are technically public even though they feel private

Non awe-related Participant Group:

The non-awe related group involved five participants who were associated with an organization that has no explicit connection to an elicitor of awe. When asked what informs their experience of awe in Cambridge during the initial interview process, members of the non-awe related group emphasized social, spatial, and temporal qualities.

Social qualities of awe in Cambridge consisted of:

- art installations that provide a 'whimsical conversation with the unknown' and point of connection with others
- institutional offerings and events, especially those that help to bring to light aspects of the city
- opportunities to meet others who benefit from the work they do
- intrigue about what is happening behind the scenes of various institutions
- being a part of the ever-changing ecosystem of Kendall Square
- intentional communities
- shared experiences with others vis-a-vis nature
- becoming a regular at a local bar or cafe and 'converting people from strangers to friends'
- people watching in parks
- the coalescence of voices and ideas at public meetings
- attending public events and festivals
- 'seeing examples of community out in the wild'
- small moments of witnessing stranger interactions and kindness

- contribution to the greater good with the knowledge that you may not reap the benefits of your contribution

Spatial aspects of awe in Cambridge included:

- an appreciation for the different squares in city and the dialogue between these squares
- the charm associated with stumbling upon small parks
- the beauty of historic neighborhoods
- the mystery of neighborhoods that have yet to form an identity
- active edges through retail and programming
- access to mobility options
- a diversity of spaces and shops to explore
- interstitial spaces
- nooks and crannies that offer brief or familiar moments of belonging

Temporal aspects of awe in involved:

- older spaces like city hall, post office, and Longfellow Bridge
- embracing variation in schedule when public transit isn't working and using this as an opportunity to purposefully slow down and enjoy new routes
- observing forces that shift and change over time and which leave behind vestiges of someone's previous life

Less common formal, ecological, and evaluative qualities of awe included:

- lighting
- a sense of adventure
- bells and whistles that add whimsy and character to a place
- messy and chaotic spaces

- 'pairing impressiveness with democracy and interconnection'
- unexpected cohesion of place
- seeing landscapes return back to life
- changes in the quality and texture of materials
- the mystery of lawn and house decorations as visual reflections of people's interiority
- things that are 'unnecessarily ornate'
- vistas across the Charles River
- the tranquility of nature

Elements that elicited a sense of awfulness included:

- routinely being confronted with a sense of terror when commuting to work by bike
- loud, busy roads
- inaccessibility
- impressiveness that aims to exclude
- the looming and foreboding quality of large buildings that feel out of context
- elite or opaque education and research institutions

Awe Walk Intervention

As part of the study, awe walks were used as an intervention with the hope that they would help to elicit experiences of awe in Cambridge. Participants were asked to go on a single awe walk over the course of a month (between the initial and follow-up interview) and record any notes and observations from this experience. In total, five out of six participants in the awe-related group completed an awe walk and five out of five participants in the non awe-related group completed a walk. Awe walks ranged from 20 minutes to 2 hours and were most commonly 30 minutes long. Participants reported taking awe walks in neighborhoods near their home, on their way to work, in familiar parks, or as part of running errands.

Across both participant groups, the awe walk intervention was reported to have positive effects on people’s well-being, with one participant explicitly mentioning that this process “brought about positive mental health benefits.” While others were less matter of fact about the awe walk experience, in general participants discussed a sense of embodiment, presence, gratitude, and appreciation. Upon reflection, one participant mentioned that they “used to just go for a walk in order to space out but with an awe walk there is more responsibility to feel my senses” while another expressed a sense of “gratitude for being alive and being able to sense things” while on an awe walk. “Rather than going from point a to b” or walking to get somewhere, one participant acknowledged the awe walk was an opportunity to walk “for the sake of experiencing the walk.” This intervention provided others with a chance to “shut off [their] mind and attune to smells” and to be “more mindful of moments of awe when they arose.” Participants expressed gratitude “for all that is out there through no work of [their] own,” a fondness for nature’s persistence, and appreciation that things are simply in the world and “free for the noticing.” In a more self-reflective manner, one participant observed how they were beginning to notice the quality of their noticing and expressed gratitude towards their inclination to “look around and experience little awes all the time.” Participants also talked about usually being distracted by personal thoughts while going on a regular walk, and how going on an awe walk helped them to feel more present, and therefore more capable of appreciating what was going on around them. Recognizing the positive impact of the awe walk process, many participants shared that they would be more inclined to intentionally seek out moments of awe. Others discussed an interest in more regularly going on awe walks. One participant observed that “if I did this more often I think it might be good for me” while another identified a clear desire to “find more time to not be rushed.”

In addition to the positive personal impacts, participants described the ways in which this intervention positively informed their relationship with their surroundings. As a result of the awe walk intervention, one participant described sensing a “welcoming feeling in the city,” a feeling of being “more impressed about what the city has to offer,” and an increased sense of connection and gratitude towards Cambridge. This same participant explained that the week following their awe walk they spent more time in libraries, they were more aware of everyday interactions, and they were more appreciative of the city's walking and bike paths. Additionally, they were more inclined to “park a little farther in order to have more time to appreciate” what was around them, more likely to walk if there was an option to do so, and that they were looking forward to spending time in their garden as an opportunity to slow down and appreciate life.

Awe Walk Insights:

Participants Reported:

- Heightened sensory awareness, presence, and embodiment
- Increased gratitude, appreciation and well-being
- Positive mental health benefits
- Increased awareness of everyday moments of awe
- Stronger connection to Cambridge through curiosity, exploration and familiarity
- Greater social and relational awareness
- Desire to seek more awe

In terms of exploration and discovery, one participant explained that the awe walk, “forced [them] to go to a new neighborhood and [see] a bunch of neat little things” and in doing so sparked an “appreciation for getting exposure to these [moments of awe] by living in a city.” Another shared that the awe walk made them “more curious to go experience other places in Cambridge” and that they felt closer to the neighborhood of their awe walk route because it provided an opportunity to build specific and granular reference points. The awe walk provided one participant with an opportunity to embrace the “scale of time and how so much of our built environment lives beyond our human lifespans.” It left one participant interested in “trying to find more familiarity in Cambridge,” and for another, it turned their focus from their personal experiences to a more future oriented perspective of what was possible in the city. Finally, it even helped one participant to “rekindle some of the joys of urban space.”

For a few participants in both groups, being prompted to think about awe was an affirmation of their existing stance towards the world. One participant acknowledged that this study reaffirmed their opinion that “there are multiple lenses to look through something. One of these can be awe and whimsy and joy and this might have dividends that aren't traditionally monetized.” Others expressed that there wouldn’t be a significant lasting impact from this study as they are already involved in the daily process of noticing and appreciating. This tendency might be explained by the fact that the selection process for participants was based on the immediacy of their responses and their relatedness to a wonder of life. Due time constraints, this study was not able to accommodate a preexisting bias towards and therefore likely reflects responses from those already inclined to experience awe in the first place.

Participants also shared that the process of being interviewed and of going on an awe walk served as a reminder, particularly “that there is awe to be found, and there are things out there to experience.” One participant shared that they are “typically inclined to look for fun neat things; but our conversation was a nudge” while another mentioned “that [study] is just a reminder to be more like my normal self rather than my running to work self.” One participant also described that “bringing my awareness of those moments of awe back through this process has made me re-fall in love with those little spaces and has reminded me of how much I enjoy those little spaces. A rekindling of the affinity for them is something I experienced.”

Though mentioning that the study wouldn’t have a lasting impact, one participant shared that the awe related conversations provided a heightened astuteness to the quality of their experiences. These conversations also gave them an opportunity to pause and reflect through which they realized “how grueling work had been.” Another participant described the ways in which the study invited more discernment about their relationship to awe, prompting them to think critically whether or not what they were experiencing was indeed awe inspiring.

Beyond personal well-being and experience of surroundings, participants in study noted ways in which the study drew their attention to their relationships with others and with their own work. One participant expressed concern at noticing how seldom they observed others around them engaged in the act of noticing. In a more connective manner, another participant shared that the lens of awe might give them an opportunity to connect more meaningfully with their child and see thing in a similar light. The study also seemed to produce an effect on participants' relationship to their awe-related and non awe-related work. Within the awe-related group, one participant described feeling inspired to take their work more seriously. Within the non awe-related group, someone expressed “wanting the human element to be more embraced at work” and wanting “to strive to incorporate awe into [their work] projects.” They also shared that as a result of thinking about awe, they were “more cognitive of the people they impact on a daily basis through [their] projects.”

Awe walk reflections from study participants:

“Cardinal song, mind settling, breathing deeply, ice becoming liquid, chickadee, red tailed hawk, squirrel sputtering a warning, grey squirrels foraging the first awakening growth in the elm branches, memories of earlier walks alone and with companions, stillness, tombstones of distant relatives a time long ago, the sound of dripping water, first thaw, a robin.”

“Although I’m familiar with this area and have walked here before... on this awe walk I really noticed what was very close to me which I usually don’t notice so much since normally I’m admiring the Boston buildings and skyline. It was interesting to see the ducks and geese swim through the liquid water but then also walk along the more solid ice. In one sense the landscape was a bit drab but also nice to pause for a moment to notice the dirty snow, emerging grass, and mud puddles since I imagine in a couple weeks the landscape will look very different.”

4.2 Quantitative Findings

In order to approach the experience of awe in Cambridge from a quantitative perspective, all instances of awe mentioned through the initial interview, follow-up interview, awe walk intervention, as well as all personal awe were recorded as spatial points and categorized based on the analytical framework of geography, morphology, phenomenology, and spatiality. General quantitative findings of the study suggest that there was no significant difference between the quantity of awe experienced by those in the awe-related and non awe-related participant group. Given the small sample size for each group and a similar average rate of awe per person per interview, instances of awe for the qualitative findings were grouped together and combined with personal instances of awe. Although Cambridge serves as the primary unit of spatial analysis for this study, all instances of awe—whether or not they occurred within city limit—were included in order to more fully capture the range of this experience. In total, 220 instances of awe were identified.

The following subsections provide a breakdown of these 220 temporalized locations of awe. To begin, overall frequencies are provided for each category to show how often each type of awe’s geographic, morphologic, phenomenological, and temporal quality appeared in the study. In order to understand which specific variable within each category most strongly shaped the experience of awe, standardized Pearson residuals were calculated for every pairing of predictor and outcome variables. Residual values above $|z| > 2$ (roughly $p < 0.05$) were treated as statistically meaningful and indicate that a particular pairing—for example, encountering nature and feeling wonder—occurred more often than would be expected by chance. For the sake of brevity, pairings for the top three variables of each category are presented and the top two associations, with $n \geq 5$, are included (refer to Appendix B for all pairings).

To understand broader patterns across categories, Multiple Correspondence Analysis (MCA) was applied to the four primary category groups, geography, morphology, phenomenology, and temporality, followed by Hierarchical Clustering on Principal Components (HCPC), helping to produce four archetypes of awe. To determine whether these archetypes were spatially concentrated rather than distributed evenly across Cambridge, global and local Moran's I statistics were calculated for each archetype using a k-nearest-neighbor spatial weighting scheme. Finally, to compare how the two participant groups and personal awe were experienced, each group was examined against the HCPC cluster typology to see whether some archetypes were more common in one than another. A chi-square test was used to determine whether these differences were statistically meaningful.

Quantitative Insights:

- Awe was most commonly encountered on the street or in parks
- Plants, architecture and art/objects were dominant forms of awe
- Experiences of awe were most often perceived through the quality of impressiveness and novelty
- Almost half of all awe were experienced through timeless features of the environment
- Awe gravitated towards four main archetypes that centered on expansive and wild nature, architecture, and human interaction
- Nature and architecture related archetypes was spatially clustered, while interactive awe was scattered
- Awe walks reliably surfaced awe with participants experiencing over 3 instances per walk

Locations

Frequencies:

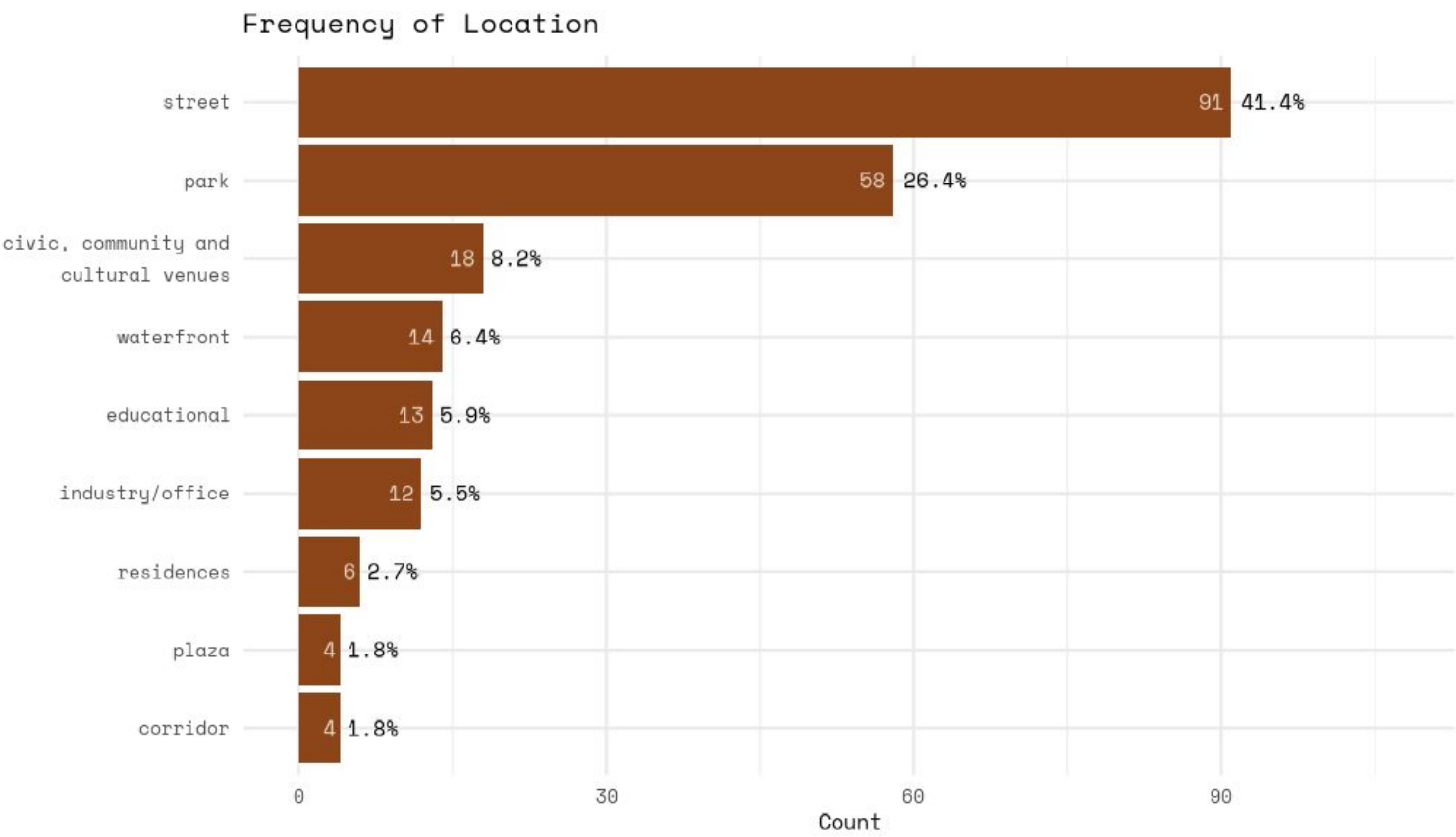
Across the nine distinct locations of awe found within the dataset, street-based experiences of awe were most common, representing 41.4% (91) occurrences. This was followed by parks which represent roughly a quarter of all occurrences at 26.4% (58). Less common was awe at civic, community, and cultural venues at 8.2% (18), waterfront at 6.4% (14), educational institutions at 5.9% (13), industry/office at 5.5% (10), residences at 2.7% (6), plazas and corridors at 1.8% (4) of occurrences.

Pairings:

Streets – Of the 91 street-based instances of awe, this location type clustered most strongly around morphologies of art/object ($z = 3.01$, $n = 21$) and architecture ($z = 3.00$, $n = 23$), and was overwhelmingly appreciated through the visual design ($z = 5.31$, $n = 38$) wonder of life. Phenomenologically, street experiences reflected the characteristic of beauty ($z = 3.75$, $n = 38$) as well as the tensions of ornateness ($z = 3.34$, $n = 17$) and novelty ($z = 2.06$, $n = 20$). Their temporal signature spanned both the periodic ($z = 2.91$, $n = 15$) and the timeless ($z = 2.34$, $n = 52$). It is worth noting that streets were under-associated with interconnection ($z=-2.32$, $n=10$) and collective effervescence ($z=-2.4$, $n=5$).

Parks – The 58 park-based instances of awe were most associated with green space ($z=4.6$, $n=10$) and plants ($z=3.23$, $n=18$) as forms of awe and, as might be expected, nature ($z=3.79$, $n=39$) as a wonder of life. The phenomenological tone of parks was that of wonder ($z=2.15$, $n=29$), wildness ($z=2.84$, $n=11$) and intimacy ($z=2.96$, $n=8$). Temporally, parks were marked by their seasonal frequency ($z=2.6$, $n=20$). Parks were also linked with the phenomenological binaries of sanctuaries ($z=2.93$, $n=10$) and thresholds ($z=2.22$, $n=8$), suggesting that they provided a sense of refuge and transition. It should be noted that parks were under-associated with impressiveness ($z=-2.04$, $n=9$), visual design ($z=-2.05$, $n=8$), and beauty ($z=-2.83$, $n=8$).

Civic, community and cultural venues – The 18 instances of awe found at civic, community, and cultural venues gravitated towards the collective effervescence ($z=4.47$, $n=8$) wonder of life and were marked by their association with communities ($z=4.35$, $n=5$) and events ($z=3.68$, $n=5$), These venues were characterized by interconnection ($z=3.65$, $n=9$), the tension of inclusivity ($z=5.44$, $n=8$), and a daily ($z=4.11$, $n=7$) temporal quality.



Forms

Frequencies:

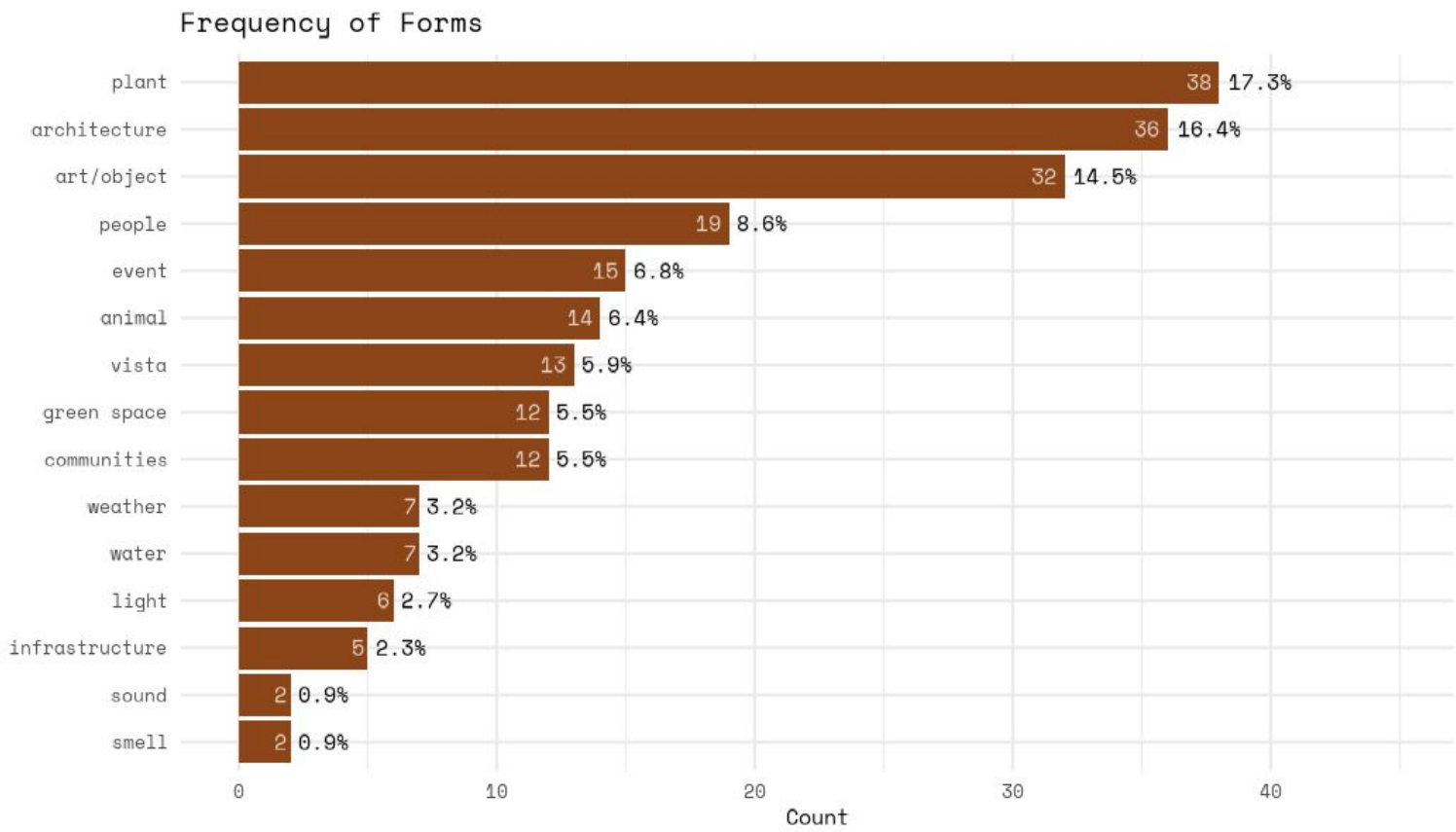
Across the 15 forms of awe identified plant-based experiences of awe were most common, representing 17.3% (38) occurrences. The second most frequent form of awe was architecture at 16.4% (36), followed by art/object at 14.5% (32), people at 8.6% (19), events at 6.8% (15), animals at 6.4% (14), vistas at 5.9% (13), communities at 5.5% (12), green spaces at 5.5% (12), water at 3.2% (7), weather at 3.2% (7), light at 2.7% (6), infrastructure at 2.3% (5), smell at 0.9% (2), and sound 0.9% (2).

Pairings:

Plants – The 38 plant-based experiences of awe clustered most strongly in parks ($z = 3.23$, $n = 18$) and unsurprisingly were over-associated with nature ($z = 7.35$, $n = 38$) as a wonder of life. Phenomenologically, they elicited wonder ($z = 3.11$, $n = 23$) as a characteristic of awe, and wildness ($z = 2.65$, $n = 23$) and impressiveness ($z = 2.33$, $n = 18$) as tensions. Their temporal signature was distinctly seasonal ($z = 4.51$, $n = 19$). Plants were also strongly tied to the phenomenological binary of alterity ($z = 5.68$, $n = 22$), or the a sense of encountering something beyond the self.

Architecture – The 36 architecture-related experiences gravitated toward industrial and office settings ($z = 4.04$, $n = 7$) as well as street-based encounters ($z = 3.00$, $n = 23$). They were overwhelmingly framed within the morphology of visual design ($z = 7.07$, $n = 25$) and most commonly produced beauty ($z = 3.99$, $n = 20$) as a characteristics of awe. Tensions of architecture revolved around the quality of being ornate ($z = 5.50$, $n = 13$) while ($z = 2.96$, $n = 8$), temporally, these encounters carried a timeless quality ($z = 5.04$, $n = 31$).

Art/objects – The art/object morphology likewise concentrated on streets ($z = 3.01$, $n = 21$) and to a lesser degree at educational institutions ($z=2.52$, $n=5$), and was almost entirely understood through the lens of visual design ($z = 6.04$, $n = 21$). Unlike architecture, which produced beauty, art/objects were more often experienced as mystery ($z = 3.53$, $n = 9$), with an ornate phenomenological character ($z = 2.28$, $n = 7$) and a timeless temporal frame ($z = 2.57$, $n = 22$). They were also linked to the phenomenological binary of surprise ($z = 2.26$, $n = 10$).



Wonders of Life

Frequencies:

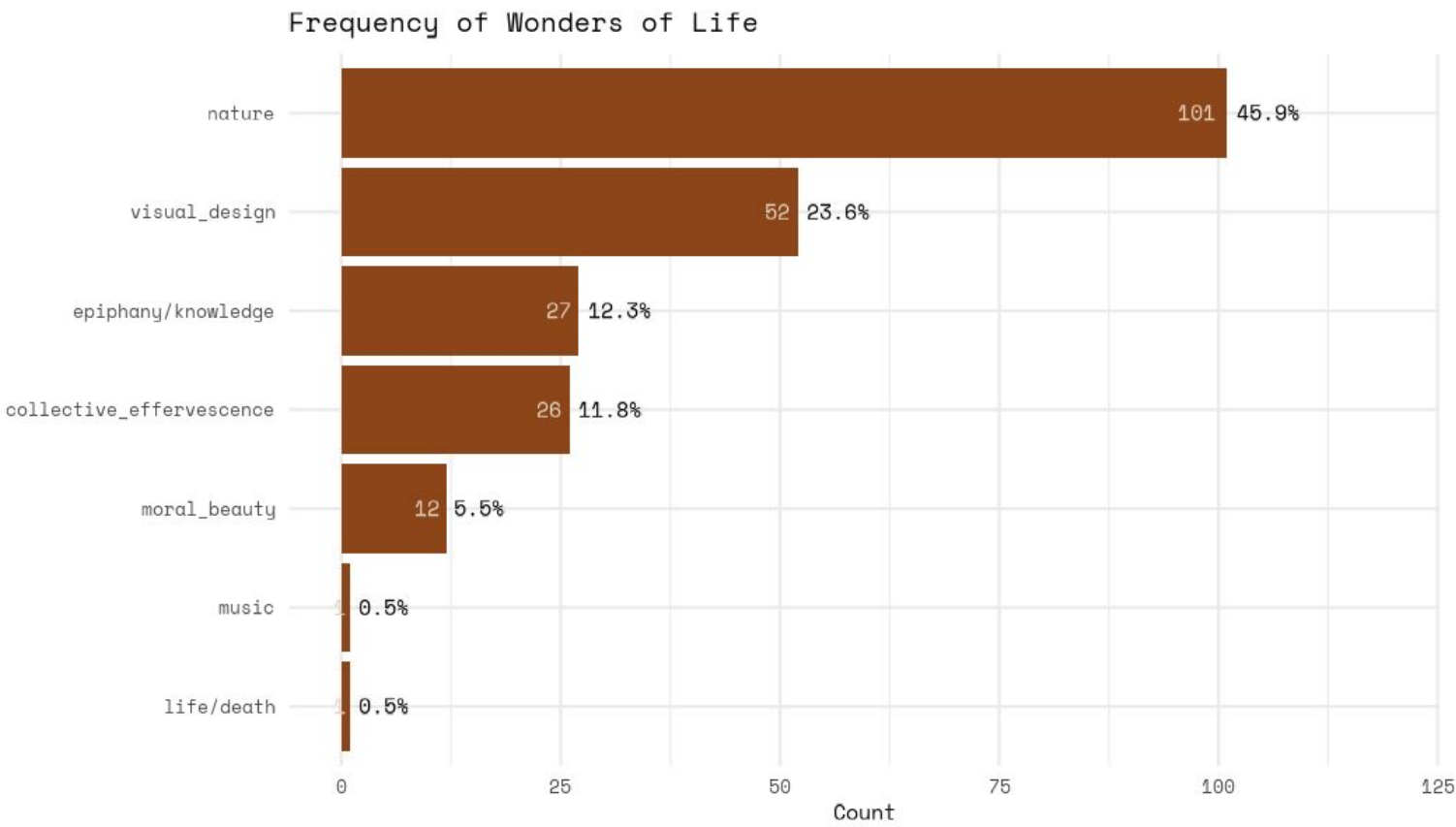
Of the eight wonders of life, nature-based experiences of awe were most common, representing 45.9% (101) occurrences. The second most frequent form of awe was architecture at 23.6% (52), followed by epiphany/knowledge at 12.3% (27), collective effervescence at 11.8% (27), moral beauty at 5.5% (12), as well as music and life & death at 0.5% (1).

Pairings:

Nature – The 101 instances of nature-based awe were mostly commonly found at both waterfronts (z=4.19, n=14) and in parks (z=3.79, n=39) and were clustered around forms of plants (z=7.35, n=38) and vistas (z=4.03, n=13). Phenomenologically, nature-based awe was experienced through wonder (z=4.57, n=55) and as something wild (z=5.23, n=21) and expansive (z=4.50, n=16). Alterity (z=5.18, n=39) was frequently a part of nature-based awe instances and from a temporal standpoint, this type of awe was seasonally elicited (z=5.04, n=39). Notably, nature was less associated with the evaluative, impressive phenomenology (z = -2.24, n = 24) than its frequency in the dataset would predict, suggesting that nature-based awe was experienced through its ecological and spatial qualities instead. Additionally, nature was under-associated with mystery (z=-2.01, n=6) and daily awe (z=-2.54, n=5).

Visual Design – The second most common wonder of life, visual design, was most often encountered on the streets of Cambridge (z=5.32, n=38), and in the form of either architecture (z=7.07, n=25) or art/objects (z=6.04, n=21). It was most often characterized by beauty (z=5.06, n=29) and the quality of being ornate (z=9.1, n=23), and was predominantly a timeless (z=7.05, n=47) frequency of awe. Interestingly, visual design was under-associated with park locations (z=-2.56, n=8), reinforcing the pattern that parks elicit awe through nature rather than their aesthetic qualities.

Epiphany/knowledge – The third most common wonder of life, with 27 instances of awe, was typically found in industrial/office (z=6.81, n=9) and educational (z=2.96, n=5) locations, and manifested most often in the form of architecture (z=2.54, n=9). These epiphany/knowledge-based moments of awe were characterized by mystery (z=2.13, n=6), their impressiveness (z=4.22, n=18), their historical quality (z=4.31, n=9), and an ephemeral frequency (z=2.96, n=5).



Characteristics

Frequencies:

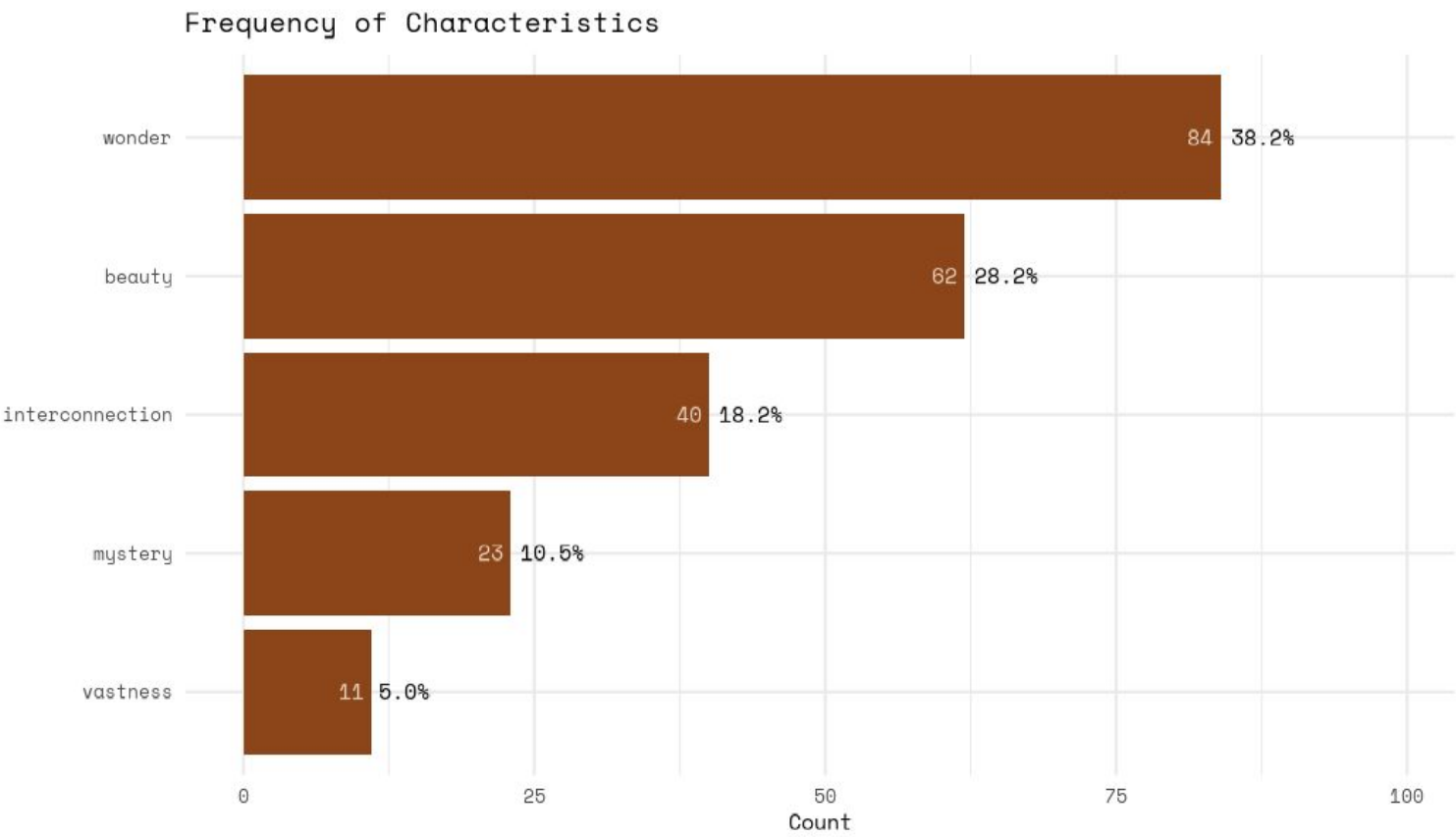
Based on data from both participant groups and personal awe, the most common characteristics of awe was wonder, representing 38.2% (84) of all occurrences. Beauty was identified in 28.2% (62) instances of awe in Cambridge, while interconnection, mystery, and vastness were experienced in 18.2% (40), 10.5% (23), and 5% (11) instances of awe, respectively.

Pairings:

Wonder – The 85 wonder-related experiences of awe were mostly associated with parks ($z = 2.15$, $n = 29$) and residences ($z=2.3$, $n=5$) and were represented by the forms of animals ($z=4.92$, $n=14$) and plants ($z=3.11$, $n=23$). This characteristic of awe was closely associated with nature ($z=4.57$, $n=55$) and was qualified by its wildness ($z=2.82$, $n=14$). Additionally, it was often accompanied with a passage through a threshold ($z=3.68$, $n=13$) and an encounter with alterity ($z=2.95$, $n=28$). Based on the criteria for these pairings, wonder wasn’t found to be significantly linked to any temporal qualities of awe.

Beauty – Characterizing 62 instances of awe, beauty was clustered on streets ($z=3.75$, $n=38$) and gravitated toward the form of architecture ($z=3.99$, $n=20$) and weather ($z=2.58$, $n=5$). Beauty corresponded to the visual design wonder of life ($z=5.06$, $n=29$) and phenomenologically it took on an ornate quality ($z=5.64$, $n=18$). Temporally, beauty was associated with timelessness ($z=2.22$, $n=37$). Notably, beauty was under-associated with the location of parks ($z=-2.40$, $n=12$) and the tension of impressiveness ($z=-2.4$, $n=12$).

Interconnection – The 40 interconnection related instances of awe were found to be clustered around locations of industry/office ($z=3.7$, $n=7$) and civic, community, and cultural venues ($z=3.65$, $n=9$). Interconnection was associated with people ($z=6.56$, $n=14$) and events ($z=3.65$, $n=8$), experienced through the quality of inclusion ($z=5.69$, $n=13$), and was most associated with a daily ($z=3.32$, $n=10$) and recurring ($z=2.96$, $n=5$) temporalities. As previously mentioned, streets ($z=-2.32$, $n=10$) were notably under-associated with instances of interconnection.



Tensions

Frequencies:

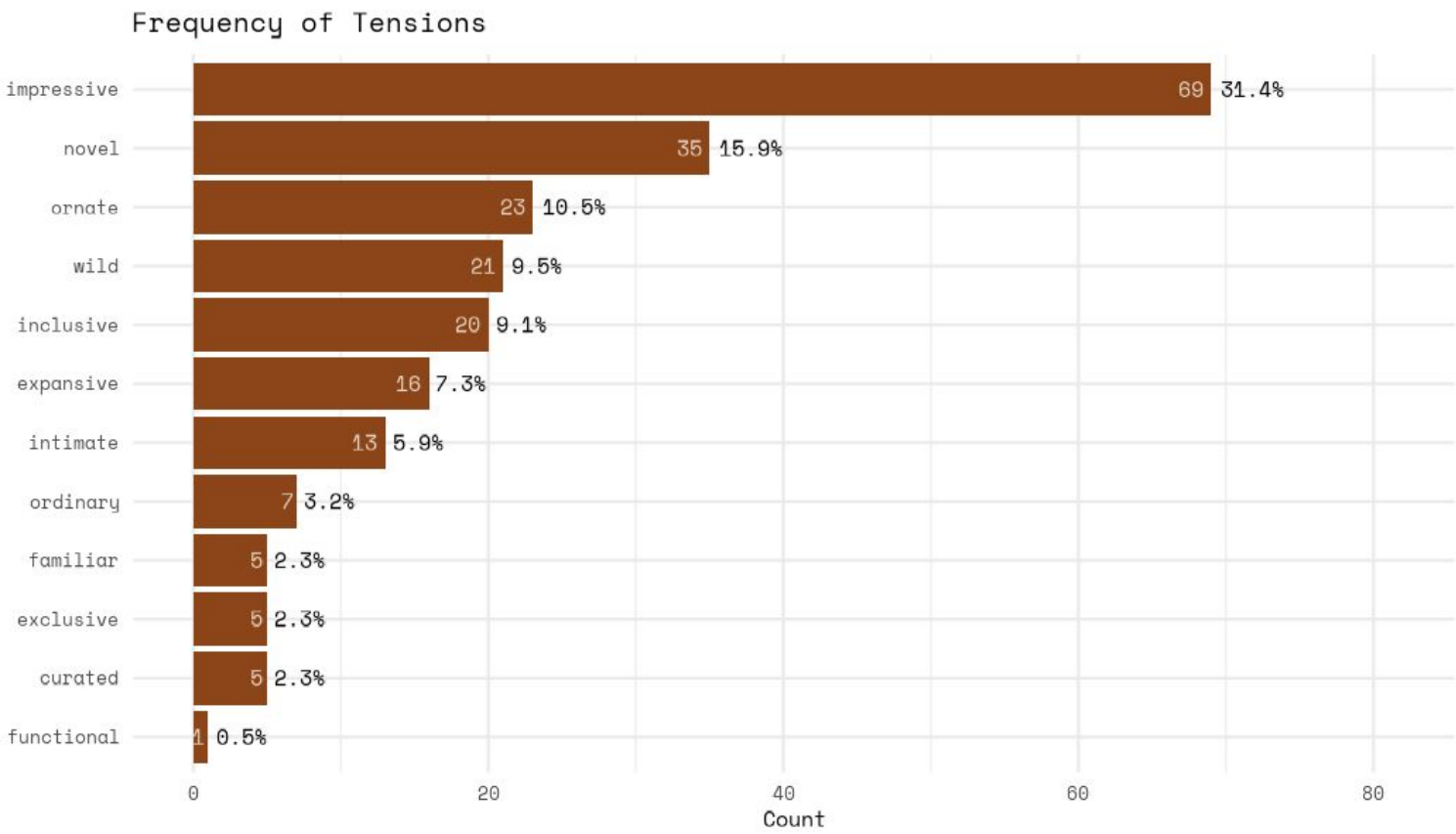
The experience of awe, as defined by the six phenomenological tensions, was most commonly found through impressiveness 31.4% (69). This was followed by experiences marked of novelty at 15.9% (35) , ornateness at 10.5% (23), wildness at 9.5% (21), inclusivity at 9.1% (20), expansiveness at 7.3% (16), intimacy at 5.9% (13), and ordinariness at 3.2% (7). 2.3% (5) of experiences of awe were categorized by the tensions of familiarity, exclusivity, and curation, while 0.5% (1) was marked by functionality.

Pairings:

Impressive – As the most common tension (31.4%, n=69), impressiveness functioned as something of a default character of awe in the dataset. Its strongest associations were with educational locations ($z = 4.27$, $n = 11$) and the epiphany/knowledge ($z = 4.22$, $n = 18$) wonder of life. Forms of awe such as events ($z = 3.05$, $n = 10$) and plants ($z = 2.34$, $n = 18$), and recurring temporal frequencies ($z = 2.33$, $n = 6$) were also over-associated with impressiveness, alongside the temporal binary of history association ($z = 2.47$, $n = 12$). As indicated in previous pairings, impressiveness was significantly under-associated with nature as a wonder of life ($z = -2.24$, $n = 24$), parks as a location ($z = -2.04$, $n = 12$), and beauty ($z = -2.40$, $n = 12$) as a characteristic of awe.

Novel – The 35 instances of novel were clustered on streets ($z=2.06$, $n=20$) and gravitated toward forms such as light ($z = 4.58$, $n = 5$) and animals ($z = 2.09$, $n = 5$). Novelty was temporally associated with periodic ($z = 3.55$, $n = 9$) and seasonal ($z = 3.19$, $n = 15$) frequencies. This tension was strongly tied to the binary of surprise ($z = 5.34$, $n = 17$), reflecting an intuitive coupling between novelty and unexpected encounter. Furthermore, novelty was associated with both periodic ($z=3.43$, $n=9$) and seasonal ($z=3.19$, $n=15$) frequencies.

Ornate – The 23 instances of the ornate phenomenology were strongly associated with the morphologies of visual design ($z = 9.11$, $n = 23$), architecture ($z = 5.50$, $n = 13$), and art/objects ($z=2.28$, $n=7$). Ornate experiences were concentrated geographically on streets ($z = 3.35$, $n = 17$) and their primary phenomenological character was beauty ($z = 5.64$, $n = 18$). The temporal register of ornate experiences was distinctly timeless ($z = 4.86$, $n = 22$).



Binaries

Frequencies:

Among the four phenomenological binaries and single temporal binary, alterity was the most common, appearing in 22.7% (50) instances of awe. Surprise was the second most common binary, found in 17.3% (38) of awe in Cambridge. History, sanctuary, and threshold represented 10% (22), 8.2% (18), and 7.3% (16) of instances of awe, respectively.

Pairings:

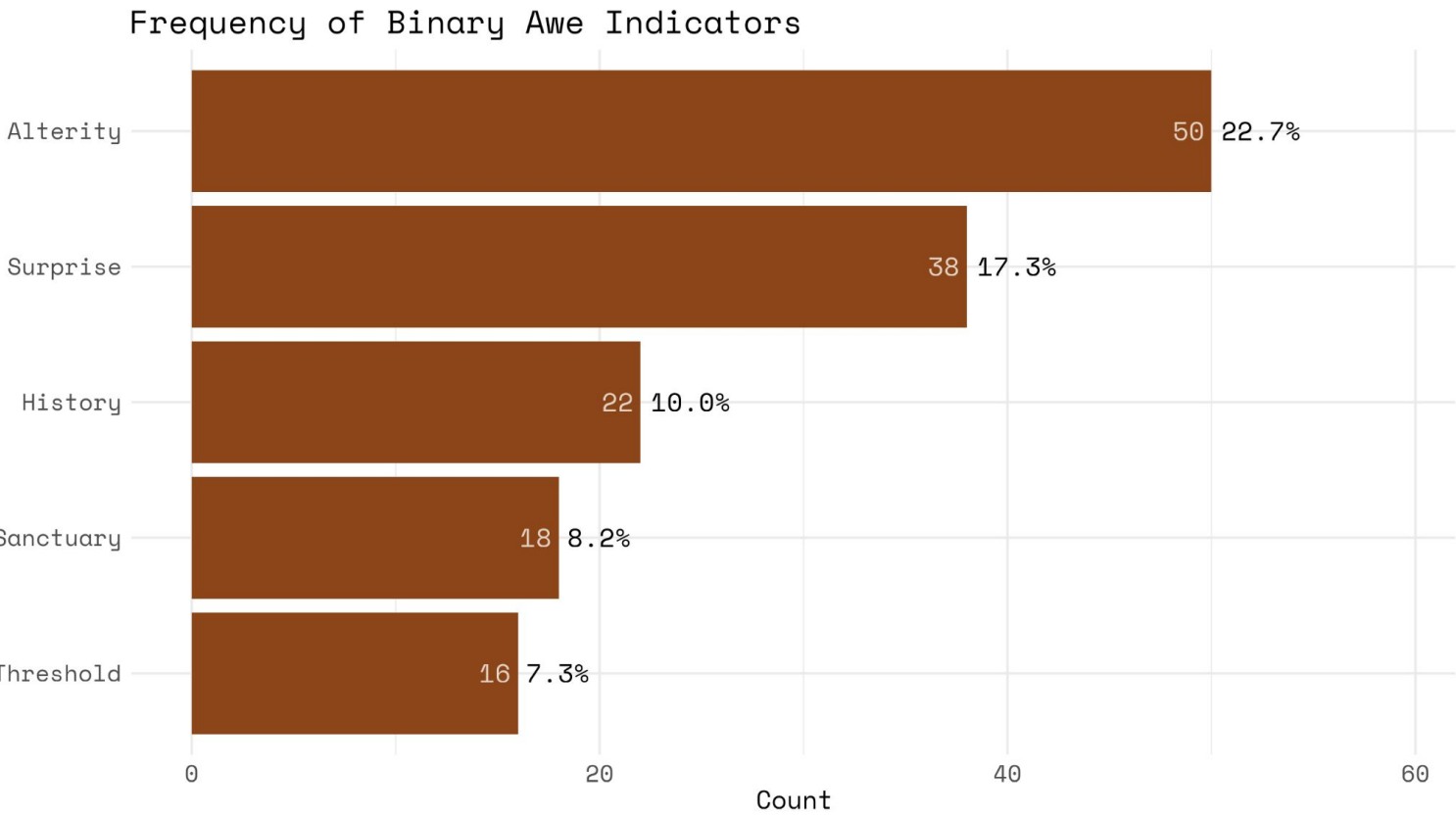
Alterity – Moments of alterity, or otherness, were clustered in parks (z=3.95, n=24) and associated with encountering animals (z=6.47, n=13) and plants (z=5.68, n=22). Befittingly, nature (z=5.18, n=39) was linked with alterity and expressed through wonder (z=2.95, n=28) and wildness (z=4.5, n=13). The frequency of otherness was commonly found through periodic (z=2.86, n=10) instances of awe. Worth noting is that alterity was under-associated with beauty (z=-2.86, n=10) and timelessness (z=-3.17, n=14).

Surprise – The 38 instances of the feeling of surprise reported in the study were most associated with animals (z=4.07, n=8) and people (z=2.36, n=7), and were most commonly expressed through mystery (z=2.34. n=8) and novelty (z=5.34, n=17). The temporal nature of surprise gravitated towards ephemeral (z=3.59, n=7) and periodic frequencies (z=2.04, n=7).

History – The temporal binary of history represented 22 instances of awe and most commonly took the form of infrastructure (z=5.27, n=4) and epiphany/knowledge (z=4.31, n=9). Phenomenologically, history was experience through the quality of impressiveness (z=2.47, n=12). Notably, nature (z = -2.99, n = 5) was significantly under-associated with history, suggesting that historical awareness in awe experiences emerged primarily from built or intellectual encounters rather than natural ones.

Threshold – The phenomenon of threshold-based awe was most commonly found in parks (z=2.22, n=8) –specifically pocket parks– and green spaces (z=5.86, n=6). It was marked by the qualities of intimacy (z=6.66, n=7), mostly likely as a result of the size of these pocket parks, and wonder (z=3.68, n=13).

Sanctuary – As a phenomenological binary, the sanctuary-based experience of awe was most commonly found in parks (z=2.93, n=10) and green spaces (z=3.26, n=4). It was marked by the qualities of mystery (z=2.5, n=5), exclusivity (n=4.27, n=3), and intimacy (z=2.02, n=3).



Frequencies

Frequencies:

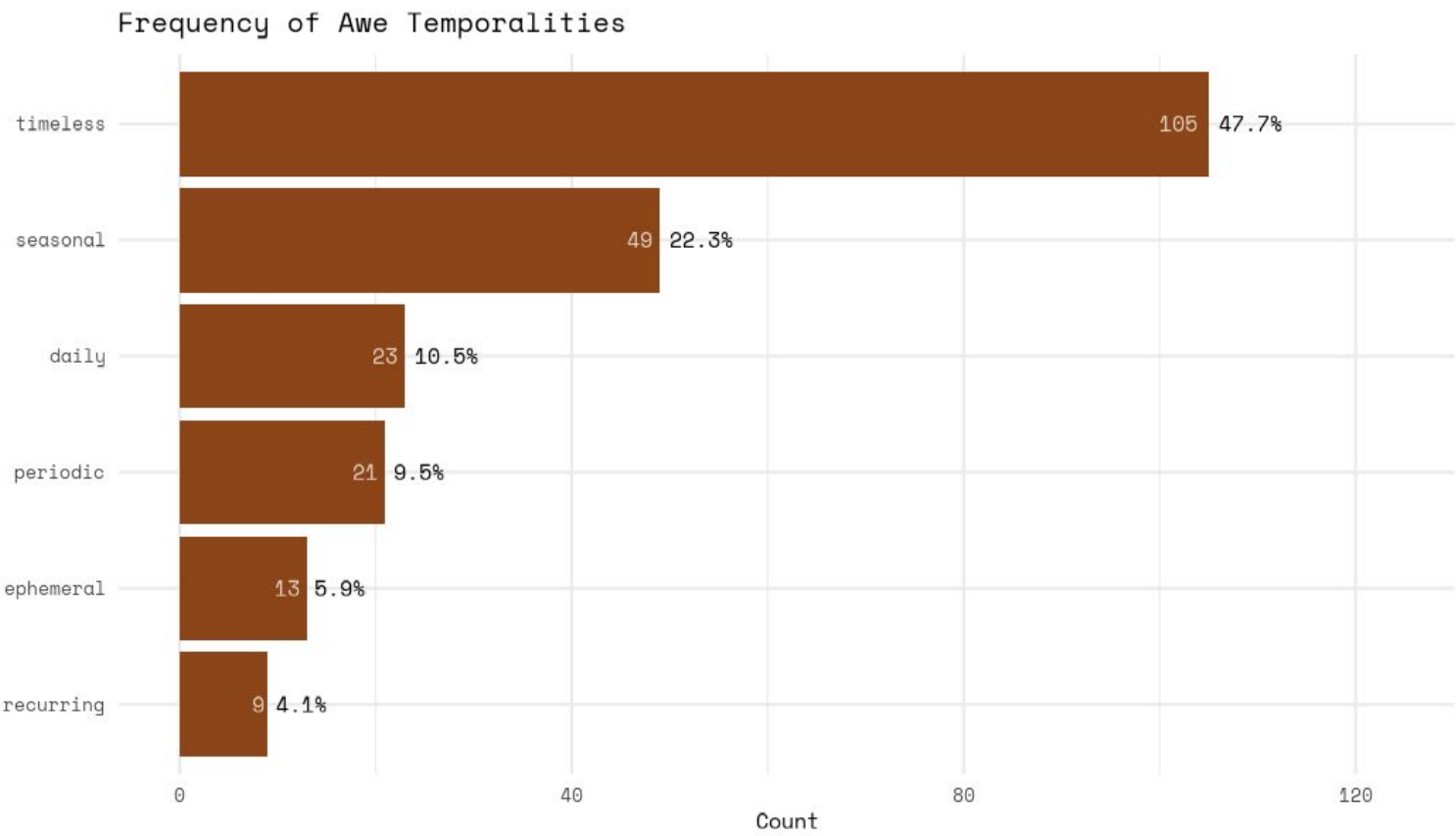
Among the six temporalities, timelessness represented nearly half of all instances of awe (47.7%, n=101). Seasonality was the second most common reflecting 22.3% (49) of instances of awe, followed by daily awe at 10.5% (23), periodic awe at 9.5% (21), ephemeral awe at 5.9% (13), and recurring awe at 4.1% (9).

Pairings:

Timeless – The 101 instances of timeless awe were clustered in waterfronts (z=2.38, n=11) and streets (z=2.34, 52). This experience consisted of forms such as architecture (z=5.04, n=31) and green spaces (z=3.13, n=11) and gravitated heavily towards the visual design (z=7.05, n=47) wonder of life. Timeless phenomenological qualities involved ornateness (z=4.86, n=22) and intimacy (z=3.13, n=12) and emphasized the characteristic of beauty (z=2.2, n=37). Worth noting is the fact that timeless was under-associated with inclusivity (z=-2.13, n=5) and wildness (z=-2.3, n=5).

Seasonal – Instances of awe that fell within a seasonal frequency were found to be most commonly associated with parks (z=2.6, n=20) and residences (z=3.64, n=5). Seasonal awe linked to plants (z=4.51, n=19), weather (z=4.09, n=6), and nature (z=5.04, n=38), and was defined by its novelty (z=3.19, n=15).

Daily – The 21 occurrences of daily awe were shaped by their presence at educational institutions (z=5.27, n=7) and civic, community, and cultural venues (z=4.11, n=11). Daily awe commonly manifested through the form of people (z=3.92, n=7) and communities (z=3.63, n=5), and as collective effervescence (z=5.65, n=11). The experience of daily awe was most associated the phenomenological qualities of interconnection (z=3.32, n=10) and inclusivity (z=3.76, n=7). Notably, daily awe was found to be under-associated with the nature wonder of life (z=-2.45, n=5).



Awe Walks

Pairings:

Of the 220 instances of awe recorded in the study, 29.5% (65) occurred while on an awe walk, 31 of which were documented by participants. This corresponded to 3.1 awe occurrences per walk, jumping to 4.6 instances per awe walk when accounting for personal awe. These moments of awe most commonly centered the forms of architecture and visual design, and predominantly took place on the street. Phenomenologically, awe walks most commonly produced the experience of impressiveness and were over-associated with the quality ornateness and beauty. They successfully produced all tensions of awe except the familiar and functional. Temporally, awe walks produced all frequencies of awe except recurring ones and were most often experienced through timelessness. Lastly, awe walk occurrences were found to be under-associated with the interconnection, likely due to the fact that these experiences were taken alone and involved a degree of solitude.

Awe Archetypes

In order to determine patterns within the categorically rich awe data, Multiple Correspondence Analysis (MCA) was used, and similar observations were subsequently grouped using Hierarchical Clustering on Principal Components (HCPC), producing a four archetypes of awe (the first 8 variables with positive v-tests are included). Moran’s I spatial autocorrelation test was then applied to determine if these archetypes were spatial clumped or if they were randomly scattered throughout the Cambridge area (refer to Appendix C for more information). Findings indicated a distinct quality of the four archetypes of awe and revealed that Clusters 1, 2, and 3 are spatially clumped, while Cluster 4 was experienced across the city without a clear geographic concentration. Ground truthing of this spatial clustering points to the fact that some of these awe ‘hotspots’ are simply the result of participants personal circumstances (i.e. where they work or live). Some of the spatial clustering, however, may help to reveal anchoring location of awe within Cambridge.

Spatialized Clusters:

Cluster One: Expansive Nature (n=19)

Components: Vista + Expansive + Waterfront + Nature + Water + Vastness + Timeless

Anchors: The Charles River Waterfront

The first cluster revealed through this analysis pointed to an experience of awe elicited by scenic encounters that evoked a sense of vastness. This type of awe involved nature and bodies of water and due to the permanence of these feature was considered temporally timeless. Spatially, it was significantly clustered (Moran’s I = 0.064, p=0.26), though more weakly due to the smaller sample size.

Cluster Two – Wild Nature (n=85)

Components: Nature + Plant + Alterity + Wild + Wonder + Seasonal + Animal + Park

Anchors: West Cambridge close to Mt. Auburn Cemetery, Fresh Pond and Danehy Park

The second cluster was the largest archetype and involved nature through a different lens. In this archetype, nature was seasonally perceived in locations such as parks through its wildness and wonder, and included encounters with plants and animals that invoked a sense of otherness. Spatially, it showed significant clustering (Moran I=0.155, p=<0.001).

Cluster Three – Built Beauty (n=74)

Components: Visual Design + Architecture + Timeless + Ornate + Art/object + Epiphany/Knowledge + Street + Beauty

Anchors: Harvard Square

The third archetype of awe centered on the beauty of the built environment. As part of this archetype, forms of awe, such as architecture and art/objects, were encountered on the street. These elements were perceived as ornate and beautiful and were timelessly available. This archetype was the second most common yet showed the greatest spatial clustering (Moran’s I=0.184, p=<0.001).

Cluster 4 – Inclusive Interaction (n=42)

Components: Collective Effervescence + Interconnection + People + Inclusive + Event + Moral Beauty + Recurring + Civic, Community and Cultural Venues

The fourth archetype of awe was characterized by recurring interactions with people through events and community spaces. This human-centered archetype was marked by perceptions of collective effervescence, moral beauty, interconnection, and inclusivity. Unlike the rest, this archetype did not show significant spatial clustering, suggesting that this kind of awe emerged through interactions around the city.

1. Expansive Nature
(n=19):

Components: Vista +
Expansive + Waterfront +
Nature + Water + Vastness
+ Timeless

2. Wild Nature
(n=85):

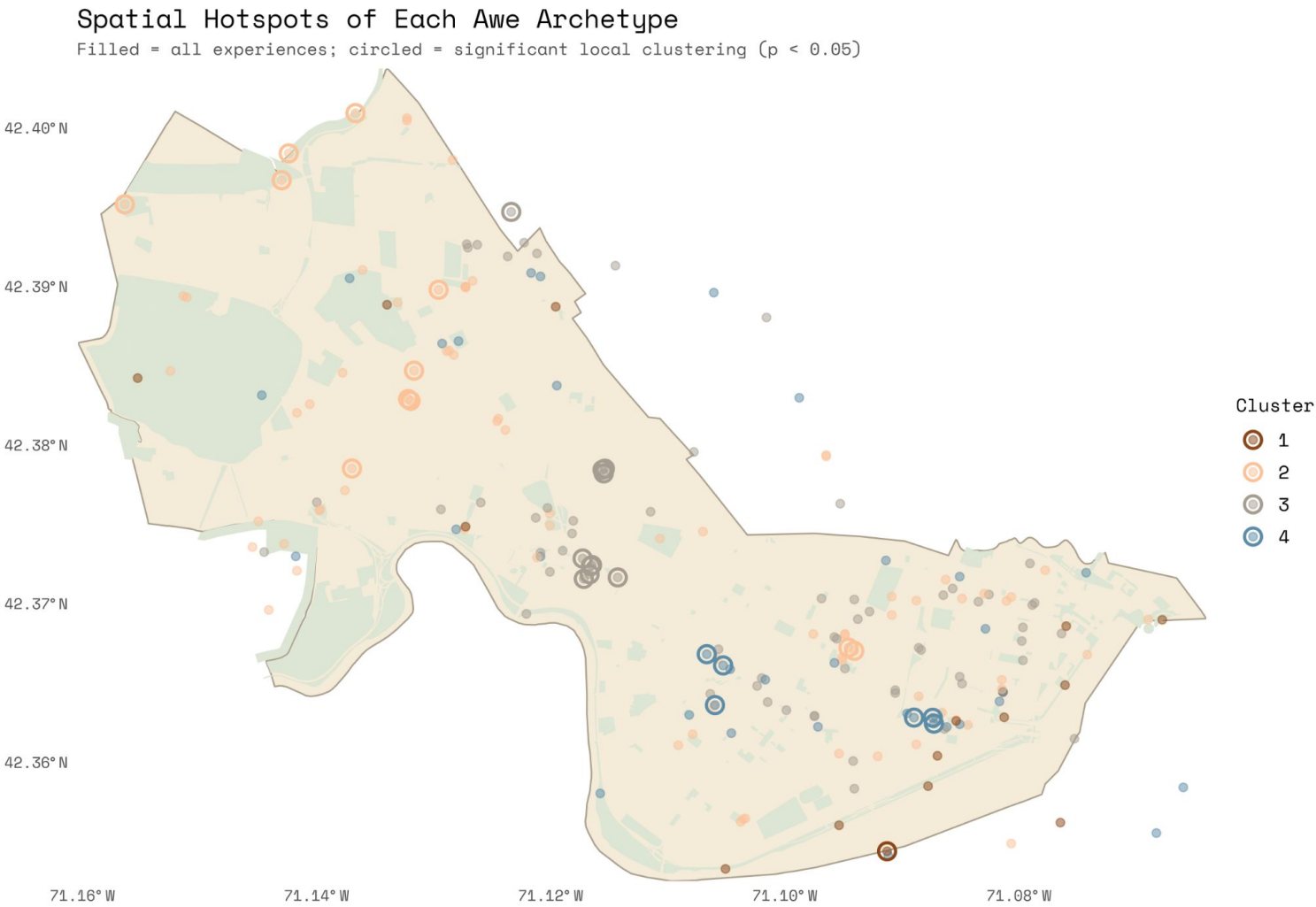
Components: Nature +
Plant + Alterity + Wild +
Wonder + Seasonal +
Animal + Park

3. Built Beauty
(n=74):

Components: Visual
Design + Architecture +
Timeless + Ornate +
Art/object +
Epiphany/Knowledge +
Street + Beauty

4. Inclusive
Interaction
(n=45):

Components: Collective
Effervescence +
Interconnection + People
+ Inclusive + Event +
Moral Beauty + Recurring
+ Civic, Community and
Cultural Venues

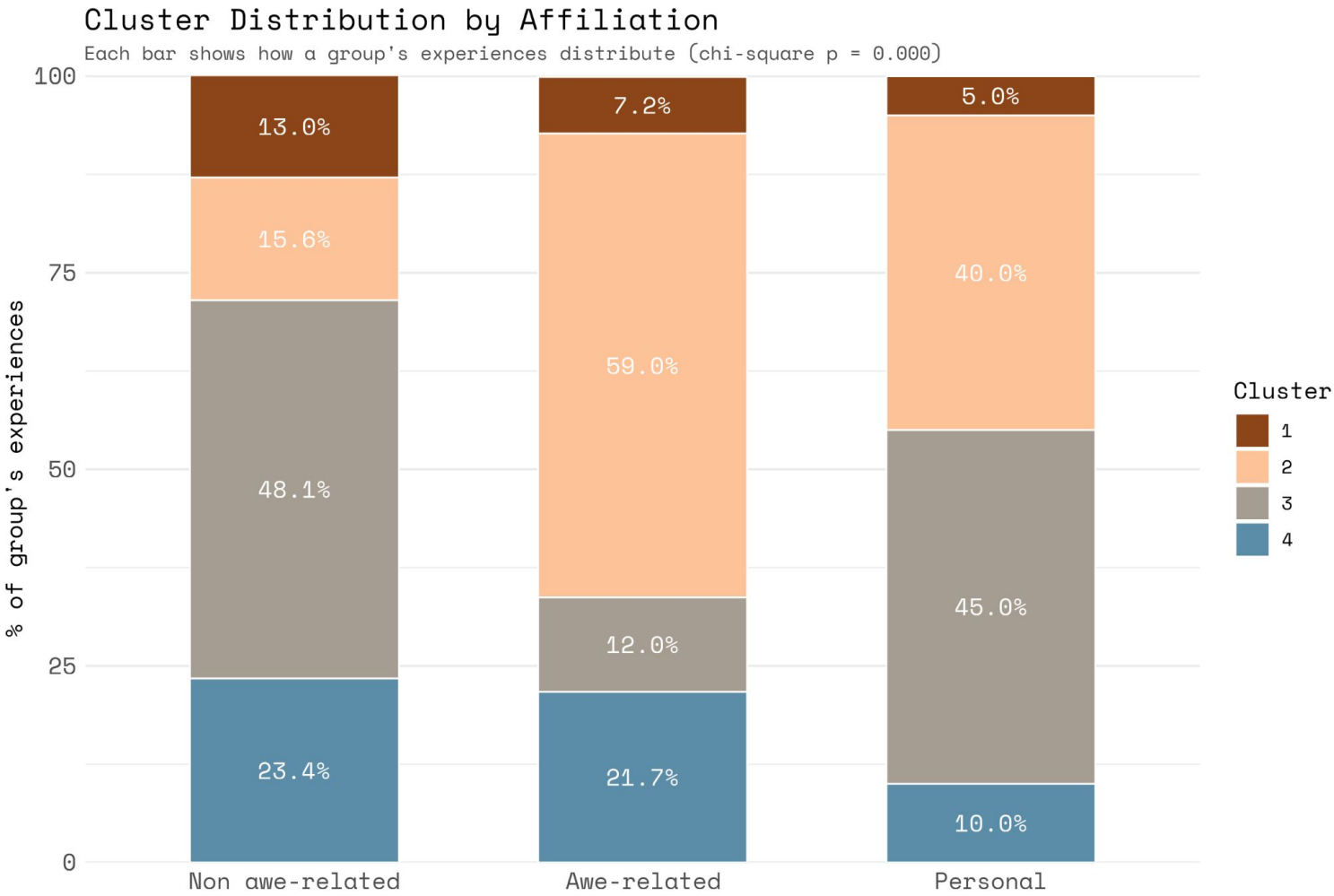


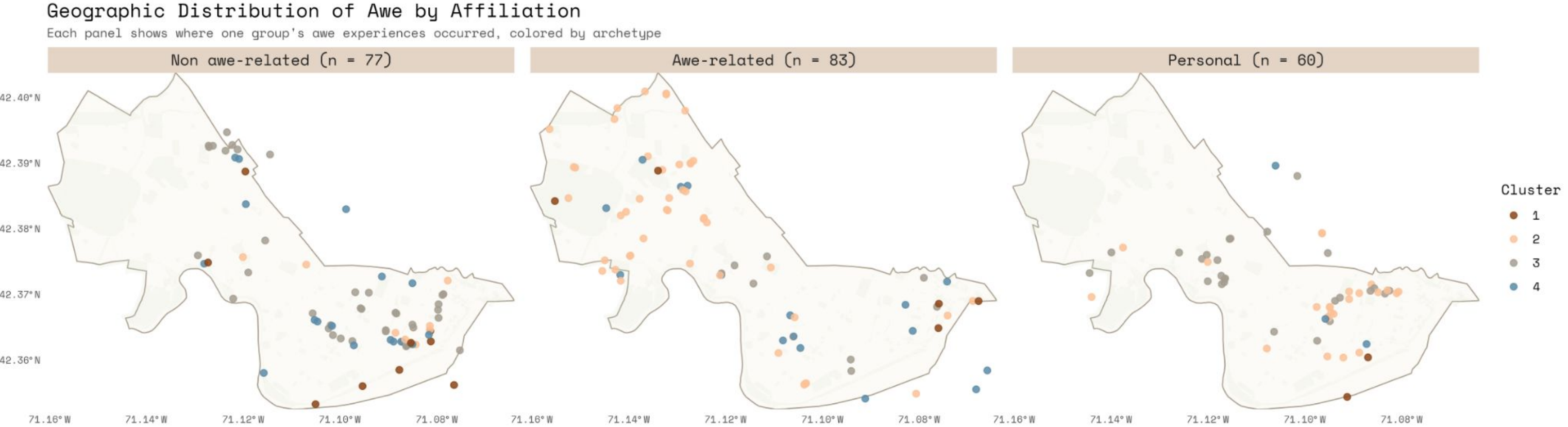
4.3 Awe by Affiliation

When asked to identify instances of awe through an initial 30 minute interview, participants in the awe-related group (n=6) reported 49 total instances, or an average of 8.2 instances per 30 minute interview. The non awe-related group (n=5) also provided 49 instances of awe during the initial interview process, which corresponds to an average of 9.8 instances of awe per person. Between both interviews, the awe-related participant group reported 83 instances of awe, while the non awe-related group reported 77.

As a means to compare the two participant groups, and to get as sense for a personal awe profile, each of these these three awe groups were examined against the MCA cluster typology. This analysis, which relied on a chi-square test to establish statistical significance, helped to answer one of the initial research questions: How do people experience awe in everyday public spaces in Cambridge, and how does this differ between groups oriented toward awe-related experiences and groups not explicitly oriented toward awe? When looking to determine if certain awe archetypes are more or less common in one group than another, results suggest that the majority of experiences (59%) for the awe-related participant group fell into the ‘Wild Nature’ archetype. Second most common for this group were experiences of Inclusive Interaction (21.7%), followed by Built Beauty (12%), and Expansive Nature (7.2%). The non awe-related participant group, meanwhile, most often encountered awe through Built Beauty (48.1%), followed by Inclusive Interaction (23.4%), Wild Nature (15.6%), and Expansive Nature (13%).

With the caveat that nature is key elicitor of awe and would therefore be present to some degree for all potential participant groups, these findings suggest that the awe-related participant groups was indeed more likely to encounter awe within their given ‘wonder of life.’ This nature-related group experienced nature-related awe 37.6% more than their non awe-related counterparts. As for personal awe experiences, the results of this affiliation comparison suggest that awe was most commonly perceived through two archetypes: Wild Nature (45%) and Built Beauty (40%). Less relevant for personal awe were experiences associated with Expansive Nature (10%) and Inclusive Interaction (5%), which was nearly absent as a archetype of awe.





4.4 An Awe Profile of Cambridge

Addressing the Research Questions:

In aggregate, the qualitative and quantitative findings described through this chapter provide a profile of how awe manifests around Cambridge. These findings not only offer a detailed perspective of awe around the city, but they also help to address, as well as inform, the initial questions guiding this research:

- How do people experience awe in everyday public spaces in Cambridge, and how does this differ between groups oriented toward awe-related experiences and groups not explicitly oriented toward awe?
- What kinds of public spaces are most frequently associated with awe?
- What urban design elements are most closely associated with participants' experience of awe?
- How do awe interventions practices shape perception of public space over time?

In attempt to respond to the first question of how people experience everyday, this study relied on an analytical framework that worked to address the notion of ‘everyday awe’ through various temporalities. This framework, informed by Rhythmanalysis as well as observations from the data itself, sheds light on the fact that awe is commonly experienced in Cambridge through timeless features of the built environment (i.e. object and things that are not limited by time), seasonal recurring experiences, and daily chronometric and diurnal rhythms. Getting to the ‘how’ part of this question involved developing an inductive framework aimed at understanding the phenomenological qualities of the awe experience. This framework, its application, and subsequent quantitative interpretation, suggest that within and around Cambridge, awe typically occurred through six distinct registers: formal, social, ecological, evaluative, temporal, and spatial. Most common among these qualities of awe were instances of connected to the evaluative, temporal, and formal registers which reflect the tensions of impressive vs. ordinary, novel vs. familiar, and ornate vs. functional, respectively. Addressing the ‘public spaces’ part of this question, which simultaneously addressed the second and third research questions, likewise required an inductive approach to identify and categorize where awe was taking place within the public realm. This process revealed that awe was mostly commonly elicited in the form of plants, architecture, and art/objects and that these moments of awe transpire frequently on the street or in a park.

Tackling the question of group differences was aided by a clustering analysis, which first provided four main archetypes of awe: 1) Expansive Nature, 2) Wild Nature, 2) Built Beauty, and 4) Inclusive Interaction. Most common of these was the Wild Nature archetype. Experienced 85 times, this pattern of awe relied on forms plants and animals, parks as the primary location, alternity and wildness as phenomenological qualities, and seasonality as the primary temporal frequency. When these archetypes were tested against the two participant groups, it became evident that experience of awe did indeed differ. While the quantity of awe instances did not vary greatly between these groups, the substance of this awe did.

Quantitative findings demonstrated that nearly 60% of awe experiences for the nature-related participant group fell into the Wild Nature archetype while only 15.6% of instances for the non awe-related group involved this type of awe. From a qualitative standpoint, these difference were corroborated in that awe-related participants reported higher levels of ecological and social awe, while non awe-related participants reported more socially and aesthetically elicited awe.

Additional Insights of Awe:

Beyond simply answering the research questions, this study sheds lights various details and factors influence how awe unfolds in Cambridge. Spatial autocorrelation of the archetypes of awe showed that three out of four are clustered. Most significantly clustered among these was the Built Beauty archetype, followed by Wild Nature, and Expansive Nature. Inclusive Interaction was found to be spatially random which can likely be attributed to fact this form of awe relies heavily on experiences with events, people, and collective effervescence, and therefore lacks a geographic focal point. Drilling down further into this spatial analysis points to a set of awe anchors for each of the archetypes. Expansive Nature was highly dependent on experiences of vistas across the Charles River by Memorial Drive. Wild Nature was found to be clustered throughout West Cambridge in close proximity to green spaces such as Mt. Auburn Cemetery, Danehy Park, Fresh Pond, and the Alewife Parkway. Built Beauty was clustered around Harvard Square and Harvard University. Though Inclusive Interaction was not determined to be spatially clustered, instances of this archetype did seem to gravitate towards Central and Kendall Square.

Pearson standard residuals analysis provided a level of granularity to help understand the relationship, or lack thereof, between factors that elicited and shaped awe in and around Cambridge. Highlights from this analysis show that streets, the most common location of awe, were often places where people experienced awe through things that were ornate or that produce a sense of novelty. Interestingly, however, streets tended to underproduce the experience of interconnection or collective effervescence. Parks, the second most common site of awe, tended to produce ample opportunities for people to experience both thresholds and sanctuary, yet in Cambridge they often lacked the quality of impressiveness and seldom elicited awe through visual design. These pairings suggest that a nature-based experience of awe around Cambridge was often associated with alterity but that this form of awe ought to produce more mystery, impressiveness, and daily occurrences than it currently did. This analysis also showed that phenomenologically the most common tension of awe, impressiveness, was overproduced by educational institutions, epiphany/knowledge, and the temporal binary of history. Novelty tended to take shape through instances of awe that included places or things made new through light or the presence of animals.

Frequencies of awe tell a different story how this emotion ebbs and flows around Cambridge. Among the six temporalities established by the analytical framework, Cambridge most often exhibited frequencies of awe that were timeless in nature. This means that awe was typically found in fixed, durable forms that don’t have a time-bound availability. The more dynamic frequency of seasonal awe was second most common and often elicited through the tension of novelty. Daily awe, which is third most common, tended to produce

the qualities of interconnection, yet failed to meaningfully invoke nature as a wonder of life.

One final, though nonetheless essential, component of this story of awe in Cambridge is the awe walk intervention. Designed as a way to get people to experience awe in between the initial and follow-up interview, this intervention, taken by 10 out of 11 of the participants, was reported as the source of numerous benefits including: heightened sensory awareness presence, and embodiment; increased gratitude, increased awareness of everyday moments of awe; stronger connection to Cambridge through a sense of curiosity, exploration and familiarity; greater social and relational awareness, and a desire to seek out more awe. This intervention achieved its desired outcome of getting people to experience awe and did so across all registers of awe. With 31 out of 160 instances of participant awe coming from these walks, it proved to be a vital source of awe within the study and seemingly reliable opportunity for participants to deepen their connection to Cambridge in the process.

Recommendations based on Cambridge’s Awe Profile:

Assessment of Cambridge’s awe profile provides several directions for how everyday experiences of awe might be fostered across the city. Given that streets are the most common location for awe yet underperform in their capacity to produce interconnection and collective effervescence, more attention should be paid to pedestrian friendly interventions that invite opportunities for meaningful or spontaneous interaction. This might include pop-up parks, street vendors, or traffic-calming measures that produce safe and welcoming conditions for connection and which simultaneously serve as a way to address experiences of awfulness reported by participants. Parks, another common location of awe, tend to be over associated with thresholds and sanctuaries yet underproduce instances of awe tied to impressiveness and visual design. To build on the strengths of parks in Cambridge, more intentional spatial boundaries, entryways, and spaces for solitude should be emphasized, and to address their weaknesses, visually appealing structures and public art pieces should be considered.

The temporal dimensions point to two further means of addressing awe. With periodic, ephemeral, and recurring forms of awe the least common, Cambridge stands to benefits from hosting more festivals, celebrations, and one-off events. Daily frequencies were shown to underproduce nature-based moment of awe, pointing to the opportunity for more small-scale interventions like green roofs on bus shelters and stormwater bioswales; medium scale interventions such sidewalk/street depaving, clustered tree plantings, and Miyawaki forests; and larger inventions like a public conservatory and botanical garden.

The four archetypes of awe–Wild Nature, Expansive Nature, Built Beauty, and Inclusive Interaction–each suggest a distinct planning possibility grounded in their characteristics. Anchoi locations that produce moments of wild nature and alterity, like Mt. Auburn Cemetery and Fresh Pond, should be conserved or expanded through intentional management practices. Existing architectural features that lend themselves to Built Beauty should be preserved while, underperforming spaces should undergo facade-improvement programs, and new construction should place more emphasis on quality and aesthetics. Places of Expansive Nature should be increased by adding new public piers, docks and bridges along the Charles River while

new lookout points should be created in places of high elevation as opportunities to acknowledge the surrounding skyline and geomorphology. Sites of Inclusive Interaction, which are currently more scattered throughout the city, could be improved by changes to street conditions or maintained by anti-displacement initiatives that preserve vital community institutions and third spaces. Finally, given the success in getting people to experience awe, stimulating their desire to find more of it, and strengthening their connection to Cambridge, awe walks and other similar awe-related interventions should be integrated into city-wide and neighborhood level programming. These inventions offer a low-cost, scalable practice that can ground people in their own contexts while also providing mental health and prosocial benefits.

Overall, these recommendations recognize that Cambridge's awe varies geographically, morphologically, phenomenologically, and temporally. Although awe in many ways relies on coincidence and deviation from the norm, and therefore resists design, the city's awe profile points to where attention and intention might best be directed. As shown, awe in Cambridge awe is not a single rhythm or elicitor to be optimized but a set of experiences and conditions to be cultivated, preserved, and connected.

Awe Recommendations for Cambridge:

- Activate streets for interconnection and collective effervescence
- Build on the strengths of parks by creating more opportunities for thresholds and sanctuaries
- Address opportunities for awe in parks finding ways to make them more impressive and visually compelling
- Host more periodic, ephemeral, and recurring activities such as festivals, celebrations, and one-off events
- Embed daily nature-based awe across the city
- Plan around the city’s archetypes of awe and preserve anchoring locations
- Integrate awe walks into city-wide and neighborhood programming

4.5 Limitations

Findings of this study ought to be read against several limitations, organized here into three groups: limitations of the sample, limitations of the data and methodology, and limitations of the framework itself.

Sample limitations:

The study relies on a sample size of 11 participants across two affiliation groups, thereby limiting the statistical comparison between groups and the generalizability of findings to Cambridge as a whole. Though demographic information was not collected, participants could generally be described as white, middle class and typically between 40-60 years old. Participants themselves were a self-selecting group of people willing to sign up for a study about awe and were most likely disposed to notice it already. This limitation was explicitly surfaced in the interview process when participants reflected on the impact of the study with many noting that they already tend to perceive the world with any eye towards awe. As such, the dataset likely reflects the perceptions of those who have cultivated an awareness of awe and not necessarily representative of a more general experience of this feeling. Collecting data from those who are less inclined to experience awe would help to establish a point of comparison and help to understand how this predisposition affects the quality or quantity of awe occurrences.

A further sample concern is the study's reliance on personal awe contributed by the researcher. While these instances of awe were necessary to expand the dataset, they introduced a known bias toward the researcher's own patterns, habits, and preferences which most often took the form of a street-based, beauty oriented experiences of awe.

Data and methodological limitations:

Data collection took place between February and March, which likely reduced the total amount of awe and skewed the distribution of awe toward winter activities and rhythms. In the follow-up interview multiple participants spoke about the challenges of getting outside for an awe walk due to the cold weather and accumulation of snow. Due to this limitation, forms that emerge primarily in summer and fall, like flowering plants and outdoor festivals—are underrepresented.

Another limitation within the data is that several instances of awe described by participants fell outside the formal boundaries of the city of Cambridge. Though these georeferenced points were retained because they reflected genuine experiences of awe, their inclusion blurs the distinction between Cambridge-based awe and a more regional understanding of awe.

Another limitation of the data is that the spatial distribution of awe is largely shaped by where participants live, work, and recreate. As a result, hotspots of awe identified through clustering analysis reflect both place-based patterns as well as the more subjective geographies of participants' routine.

As a result, it is difficult to determine if underrepresented neighborhoods in Cambridge demonstrate a meaningful absence of conditions for awe or simply an absence of participants who passed through these locations.

Given that awe is subjective, classifying it without a discrete and fixed phenomenological reference point introduces uncertainty at every stage of the analysis. The coding of phenomenological tensions and binaries, morphological forms and wonders of life, and frequencies relied on the researcher's interpretation of participant descriptions. While this reading was informed by the associated conceptual frameworks as well as participant language, and refined through iterative coding, another researcher applying the same framework would yield different results.

The four archetypes of awe generated through clustering analysis reflected limitations of the sample size and analytical framework and therefore ought to be treated as a preliminary set of descriptors. With more data, additional archetypes might emerge, and the existing four might be further clarified or redefined altogether. Likewise, the spatial autocorrelation analysis which identified hotspots and anchors may only be true relative to the current dataset. As with all of the quantitative methods, the standardized Pearson residuals analysis, which was used to identify over- and under-produced relationships, could yield further insight with a larger, more robust dataset.

An unanticipated limitation of the study was that the interview process itself functioned as an awe intervention. Independent of the awe walk, multiple participants reported that the initial conversation heightened their attention of awe, meaning that data collected from the follow-up interview was informed by both the awe walk and the initial conversation. As a result, the 160 participant instances of awe measure both unprepared and cultivated descriptions of awe. It is also worth noting that the 31 participant awe-walk instances follow a different protocol than the initial interview process in that participants were primed to think about awe and direct their attention towards it. Due to the nature of this intervention, awe walk instances are typically more aesthetic and street-based than instances reported from memory. Although seldom distinguished throughout the analysis, instances of awe derived from the initial interview, follow-up interview, and awe walk intervention reflect methodological variations.

Framework limitations:

A key framework limitation of this study is the challenge associated with defining awe. During the outreach process, participants were told that awe could be understood broadly as an experience of wonder, mystery, or a sense of something larger than oneself. However, given that awe is a complex emotion closely associated with surprise, gratitude and reverence, and given that participants came with preconception about what awe meant to them, it is reasonable to assume the exact definition of awe varied for participants. The study's reliance on self-reported versions of awe meant that what was coded as awe came down to what participants recalled or experienced given their current understanding of the emotion. Many shared a tendency to find awe in grandness rather than the everyday, which may also point to the need for a definition of awe that relies less on vastness and more on deviance from the norm.

Cantabrigian Awe

An additional framework limitation is the lack of a clear definition of what is considered awful. This form of threat-based awe was acknowledged by a handful of participants, yet there wasn't a framework to understand and interpret flavors or variations of awful. As such, the study did not explicitly inquire about moments of awfulness. Future research could benefit from an investigation that compares awe-inspiring and awful moments. A greater understanding of the impact of awfulness could also determine how the benefits of awe-inspiring moments are offset or counteracted by awful ones. Analytically, an understanding of awfulness may also help to identify spatial hotspots and awful archetypes that ought to be addressed and remediated.

Section Five



Implications for Planning

In cities...people report more robust health, even after controlling for income and local levels of pollution. In cities with pathways for walking, orienting landmarks, squares, and public buildings like libraries—elements of urban design that locate us within the geometries of social urban living—people feel more open and report great health and wellbeing.

- Dacher Keltner, 2023, *Awe*, p. 184

5.1 Cities of Awe

What does it mean to design for awe? Can this fleeting emotion be captured and tamed or must it be coaxed out and welcomed? What would a landscape rich in awe look like, and what would this abundance of awe say about its inhabitants and their relationship to this landscape? In what ways is a drive for awe reflective of a larger desire for cities and spaces to feel a certain way, rather look or perform a certain way? What might this drive for a city with an abundance ask of us and what are we willing to give to make it happen?

The hope is that this study and its accompanying framework gives practitioners a foothold to answer these questions. The work of designing for awe is no simple task. If we want awe to become a feature of urban life it will require a willingness to rethink our personal and collective relationship to time and space as well as our relationship to awareness and emotion. Given the ways in which awe manifests across different temporal registers and how it has been found to impact people’s perception of time, designing for awe will involve an integrative approach to time and space. Such an approach will meaningfully address Rhythmanalysis notions of “temporalized space” and “localized time” by honoring the multitude of temporalities—past, present, and future—embodied by human and non-human others in a given space. It will involve a greater appreciation of a sacred form of history—one that is active, lived, and celebrated—and may substantiate a further investigation of the ways in which places of awe alter individual and collective sense of time. Designing for awe will also mean navigating perceptions and prior experiences of awe. As became evident over the course of conducting this study and by various conversations about the study itself, awe is often regarded as something remarkable and is less commonly seen as a feature of everyday life. Although we certainly deserve experiences of awe like beholding the Grand Canyon or witnessing a solar eclipse, we are more likely to encounter, though perhaps less likely to notice, the unassuming experiences of awe like observing a stranger’s kindness or hearing the daily chatter of a host of sparrows. Learning to differentiate between the capital “A” forms of awe that tend to be jaw dropping, transcendent, and exceedingly evident and the lowercase “a” forms of awe that are subtle, interstitial, and unobtrusive will mean developing a greater attunement to and appreciation for the rhythms of life, no matter how vast or seemingly insignificant they might be. On a larger scale, this attunement could be achieved through educational campaigns and programming that help people to learn about the benefits of awe and that promote slowness and mindfulness as preconditions for awe. Importantly, education and programming would be oriented towards helping people to identify awe within themselves. As highlighted in this study, awe has a self reinforcing quality and can be brought to life simply by talking about it or looking for it. However, people first need to know what they are feeling internally, and must have the language to describe this inner experience. This ability to precisely identify and label emotions is known as emotional granularity and is associated with lower rates of anxiety and depression, increased emotional regulation (Kashdan et al., 2015), and great resilience through adaptive coping strategies (Tugade & Fredrickson, 2004). Designing for awe is therefore not simply a spatial project, but a cultural and emotional one that asks how urban life might better support attention, connection, meaning and emotional awareness. ab

In an attempt to design for awe, this study suggests some promising starting points. Experiences of awe can be classified and analyzed both categorically and spatially. Classifying awe may look like identifying geographies (where awe takes place), morphologies (what forms elicit awe), phenomenologies (what is the quality of the experience of awe), and temporalities (how often can this experience of awe recur). These four categories and their constituent variables offer a way to hone in on an experience of awe and determine its relative uniqueness or replicability. In aggregate, these experiences of awe can point to larger trends, or archetypes, that qualify the nature of awe in a given neighborhood or city. These patterns of awe can be further analyzed to understand where and to what degrees they are spatially clustered.

Beyond just designing *for* awe, practitioners must begin to question rational, technical models to city planning by considering how to design *with* awe. Rather than taking an extractive approach—one which relies on reproducing as much awe as possible in order to derive as much benefit as possible—we ought to consider building mutual and reciprocal relationships with the rhythms and conditions that help to elicit awe. First and foremost, designing with awe will mean increasing our sensitivity to this emotional state so that we know it when we feel it. It will involve a willingness and opportunity to slow down and attune to our embodied experiences, sensory perceptions, and personal rhythms so that we can be open to unexpected encounters and to the affective quality of our environments. However, given that there is no guarantee of feeling awe even when we are open to the experience, the best we can do is set ourselves up for success. As such, designing with awe will mean creating the conditions, or preserving existing ones, that can give rise to moments of awe. This will require providing ample opportunities for interconnection and moral beauty as well as spatial conditions like expansiveness and intimacy. It will involve an attention to detail and a willingness to invest in unnecessarily beautiful things, even if these may occasionally go unnoticed. Due to the ways in which nature is a prime elicitor of awe, creating conditions for this emotional experience will be an ecological project. More parks, trees, and green spaces will increase the likelihood of experiences of awe by inviting the rhythms of wildlife into cities. Alternatively, promoting ecological awe may be achieved creating spaces outside of cities for the rhythms of nature to flourish. This would mean upholding and re-establishing clear spatial boundaries for cities, thus providing spaces for encounters of otherness as well as opportunities to pass through an ecological-urban threshold. In addition to creating conditions for awe, we must work to honor and preserve spaces and things that are existing wellsprings of awe. This may involve a more animate form of history that celebrates decisions in the past to preserve green spaces or waterfronts, limiting encroachment on culturally and ecologically significant places, and enforcing historic preservation requirements despite current architectural trends.

Ultimately, moving from designing *for* awe to designing *with* awe will mean releasing the potential of rhythms and circumstances by attuning to what these conditions of awe ask of us. It will mean creating and sustaining relationships based on reciprocity so that the magic of awe can surface. It will mean seeing cities as emotional spaces, defined by how we want them to feel, not just how they function or what they can produce. It will mean working with our innately human capacities and embracing the potential for walking as practice “for a radical placement and displacement of self, fixing and unfixing self to urban structure,

Implications for Planning

locational politics and cultural form” (Labelle, 2008 as cited in Edensor, 2010, p. 7). At its broadest, an awe-based approach to planning substantiates the need for an urban ethic.

Building upon Aldo Leopold’s concept of a ‘land ethic’ (Aldo Leopold Foundation, n.d.), a conviction that emphasizes the moral responsibility humans have to respect and care for biotic communities, an urban ethic fit for awe calls us to take our responsibility to the systems, spaces, and communities that constitute urban life. Informed by Keltner’s understanding that “awe enables us to see the systems underlying the wonders of life and locate ourselves in relation to them” (Keltner, 2023, p. 256), an urban ethic asks us to take responsibility for these systems so that they retain their capacity for diverse rhythms and their potential for sacredness. By embracing an urban ethic we stand to lose the sense of stability and order we’ve grown accustomed to, yet what do we stand to gain? In disrupting the monotony of everyday life, we open the possibility of a different kind of city: a city of awe – one marked by slowness, simplicity, spontaneity, and emotion, where the subtle moments of beauty, wonder, mystery, interconnection, and vastness buried just beneath the surface can finally come into view.

5.2 Finding Rhythm

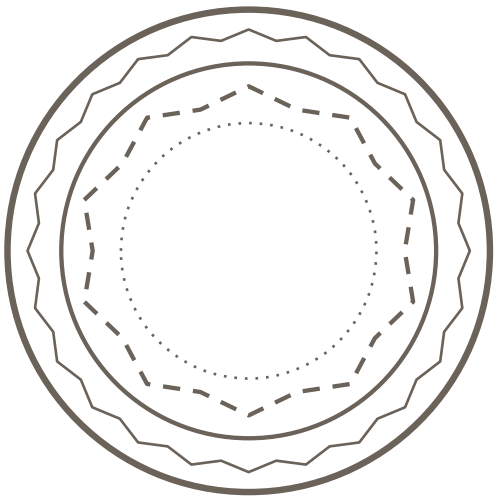
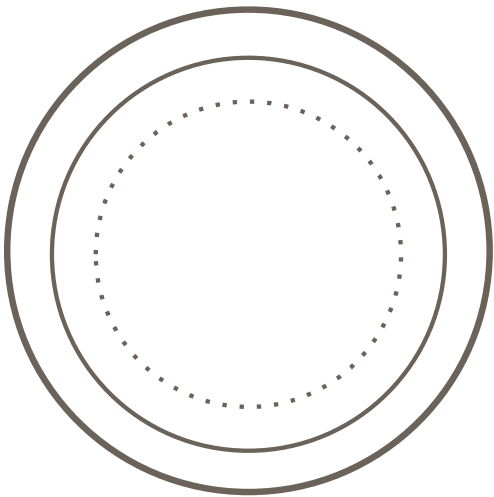
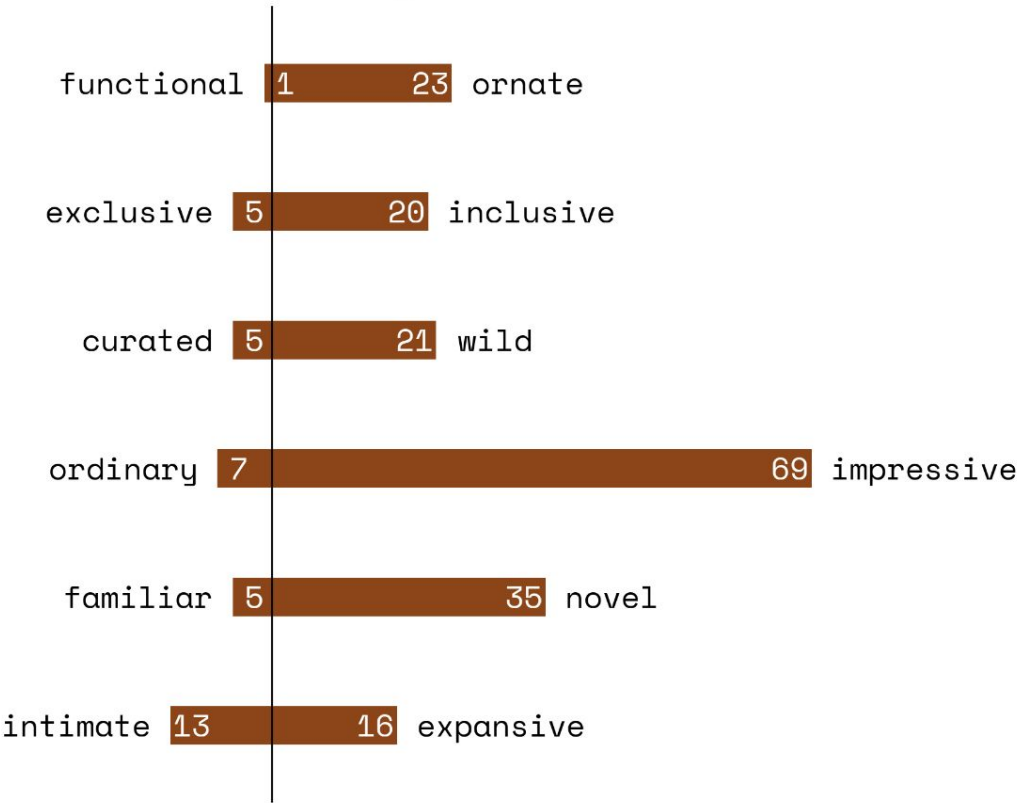
Cities are inherently polyrhythmic. Everyday urban life is composed of countless overlapping rhythms, some of which are more dynamic like diurnal patterns, weekly routines, and seasonal cycles, and some that are more stable like development, evolution, and geomorphic features. Due to the mundane and repetitive nature of everyday life as well as the slowness of more stable forms, many of these rhythms go unnoticed. Therefore, beyond simply adding new rhythms to this preexisting assemblage, awe-based planning should focus on revealing, refreshing, and reawakening the rhythms that are already present. As Keltner observes, "awe arises when we perceive change” (2023, p. 24) and through its pairing of time and space, change is what rhythmanalysis helps to make perceptible. With the help of rhythmanalysis, the task of eliciting awe is activation rather than creation—a process that relies on shifting existing rhythms and their relationships to one another.

Arriving at an understanding of how experiences of awe can benefit from subtle or significant variations in their constituent qualities came through observing and considering the tensions of awe. As participants described their experiences of awe and as these experiences were classified through the analytical framework, six phenomenological tensions began to take shape: ornate and functional, inclusive and exclusive, wild and curated, impressive and ordinary, novel and familiar, and expansive and intimate. Each tension explored a different register through which awe manifested including the formal, social, ecological, evaluative, temporal, and spatial. Together, these tensions and their corresponding registers offered a way to describe the quality or kind of awe that was routinely surfacing. What became clear, however, was that not only were these tensions expressed to varying degrees, but they also carried two distinct qualities that could be mapped onto Lefebvre’s concepts of arrhythmia (a state of disruption and discordance) and eurythmia (a state of alignment and equilibrium). Some tensions, like impressiveness or expansiveness, depended on disruption, transformation, unboundedness, and chaos, while others, like familiarity or intimacy, depended on harmony, continuity, constraint, and alignment. Thus, through the lens of rhythmanalysis, arrhythmic tensions of awe can be defined by renewal, unexpected encounters, things outside of the norm, or instances that break routine. Eurythmic tensions of awe, meanwhile, reflect repetition, slow recognition of beauty, a deepening familiarity, the rhythms of care, and details that come into view only when we slow down enough to perceive them. Importantly, both rhythms of awe arise from change: arrhythmic awe suggests a more external change often set against a stable, everyday, or normative backdrop whereas eurythmic awe suggests more subtle or internal changes, particularly as a rhythm or set of rhythms coming into alignment. Though the notions of arrhythmic and eurythmic awe help to further ground this research within the framework of rhythmanalysis, they are ultimately less a way of categorizing tensions than a way of confronting their essential qualities. They name two modes through which awe becomes perceptible and offer a lens that can be carried across the analytical framework as a whole. Rhythmically understanding a place, however, involves listening to its constituent parts—its geographies, morphologies, phenomenologies, and temporalities—regardless of whether awe is arising arrhythmically through disruption or eurhythmically through alignment.

Implications for Planning

The work of operationalizing rhythmanalysis within the context of awe-based planning mirrors the very modes through which awe arises. Rhythmanalysis becomes the eurythmic process of seeing a place for what it is. This consists of identifying what rhythms are present, appreciating the composition of dynamic and stable forms, and acknowledging what tensions and temporalities they carry. Analyzing rhythms in this way operationalizes the conceptual framework underlying this study by tracing the geographies, morphologies, phenomenologies, and temporalities through which a place currently produces moments of awe. Rhythmactivation, by contrast, looks at sites potential for producing awe and is the more arrhythmic work of reimagining, renewing, and intervening. It involves introducing new rhythms and rhythm relationships, refreshing dormant ones, and accentuating tensions in ways that highlight the subtleties or significance of both eurythmic and arrhythmic awe. This activation of rhythms provides an opportunity to explore the interplay between vastness and accommodation and to address the ways in which “the novelty of a stimulus, and eventual habituation to it, are likely critical determinants of awe” (Piff et al., 2015, p. 896). Moreover, this practice of reimagining a place offers suggestions for how a site's rhythmic profile might be emphasized based on its strengths and how this profile might be amplified, reset, or rebalanced based on its latent potentialities. The following section provides an example of a rhythmanalysis and subsequent rhythmactivation for the Greene-Rose Heritage Park in Cambridge, MA. Though subjective in nature, these comparative diagrams demonstrate how rhythmanalysis and the analytical frameworks of this study can move from theory to practice.

Pairings of Phenomenological Tensions



Eurythmia
+
Repetition

- harmony
- continuity
- constraint
- alignment

Heuristic: Is this rhythm reinforcing, reiterating, or controlling?

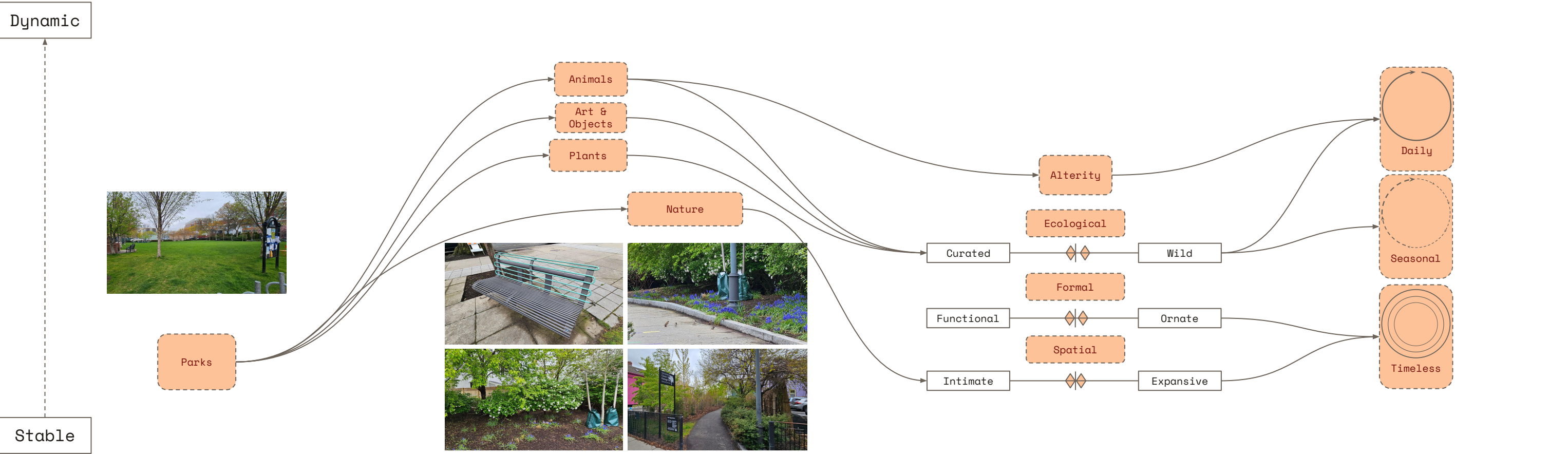
Arrhythmia
+
Renewal

- disruption
- transformation
- unbounded
- chaos

Heuristic: Is this rhythm undoing, changing, or interrupting?

Rhythmanalysis of the Greene-Rose Heritage Park

A snapshot of the awe potential at the Green-Rose Heritage Park as of April 30th, 2026.



This rhythmanalysis indicates that currently Greene-Rose Heritage Park incorporates forms like art/objects and plants yet underperforms in its ability to move the tensions related to these forms towards the ornate or functional side of the spectrum. This results in an experience that produces few moments of intimacy, and a neutral blend of curated and wild ecology. However, due to the on-site Miyawaki Forest and near-daily chorus of sparrow that occurs in the bushes adjacent to this forest, the park does lends itself to the experience of alterity. Temporally, the park is animated by seasonal changes that invite wildness and beauty, daily frequencies that involve alterity, and timeless frequencies to its fixed forms.

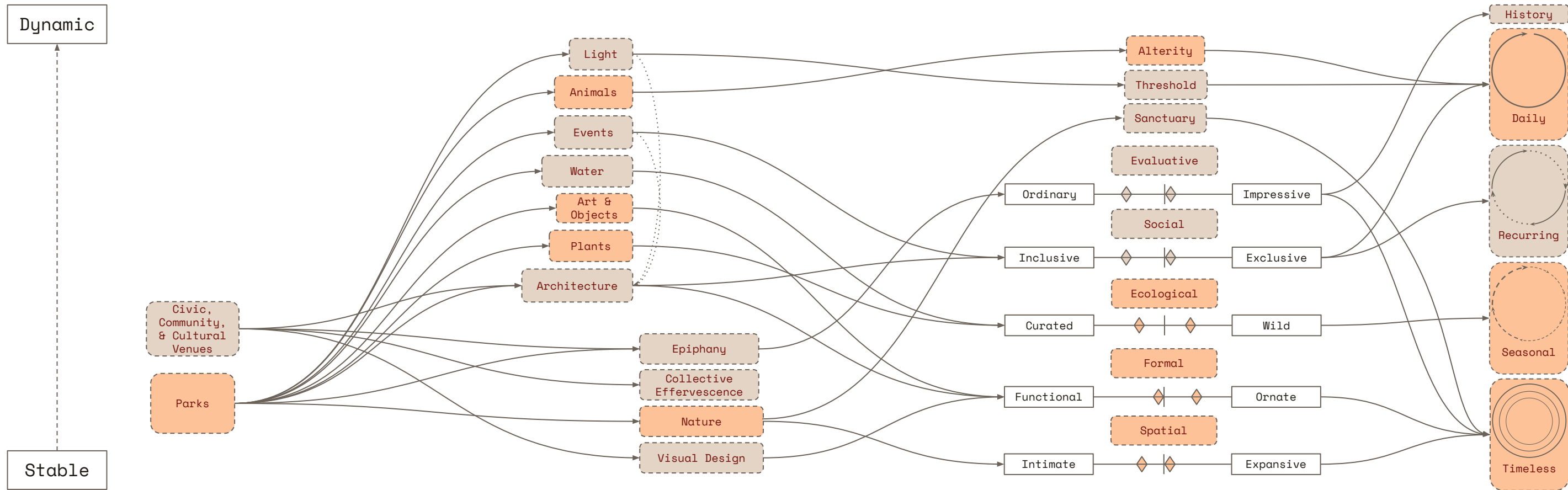
Geographies	Morphologies		Phenomenologies	Temporalities
Locations Civic, Community, & Cultural Venues Corridors Educational Institutions Industry & Office Buildings Parks Plazas Residences Streets Waterfronts	Forms Animals Architecture Art & Objects Communities Events Green Space Infrastructure Light People Plants Smells Sounds Vistas Water Weather	Wonders of Life Moral Beauty Collective Effervescence Nature Music Visual Design Spirituality/Religion Life & Death Epiphany	Tensions Formal Ornate ↔ Functional Social Inclusive ↔ Exclusive Ecological Wild ↔ Curated Evaluative Impressive ↔ Ordinary Temporal Novel ↔ Familiar Spatial Expansive ↔ Intimate Binaries Alterity Surprise Threshold Sanctuary	History Ephemeral Periodic Recurring Seasonal Daily Timeless

Implications for Planning

Rhythmactivation of the Greene-Rose Heritage Park

Suggestions for how to increase awe and accentuate rhythms

Activation of the rhythm of the Green-Rose Heritage could increase the potential for awe by introducing a new location type, adding forms that allow for tensions, and accentuating current tensions. Providing an opportunity for commerce through the introduction of a small cafe or daily food truck could create opportunities for collective effervescence and inclusion. Adding new forms, especially stables one like a visually appealing gazebo or water feature could help to balance the more dynamic ones like plants and animals. Forms like nightly lighting of the walkway between the Miyawaki Forest could establish a more defined threshold, or entryway, for the park. Accentuating current tensions of wildness by expanding the Miyawaki Forest could increase the potential for alterity. Adding new gardens with an array of flowering plants and shrubs could bolster the curated tension while also providing spaces that provide solitude, intimacy, and sanctuary. In efforts to address the temporality of this site, the history behind the name and its heritage could be included through signage and celebration. These elements would also encourage opportunities for epiphanies about the origins of this location. Furthermore, recurring events like weekly exercise classes or Miyawaki Forest field trips could create potential for recurring frequencies.



Geographies

Locations

Civic, Community, & Cultural Venues	Corridors	Educational Institutions
Industry & Office Buildings	Parks	Plazas
Residences	Streets	Waterfronts

Morphologies

Forms			Wonders of Life	
Animals	Architecture	Art & Objects	Moral Beauty	Collective Effervescence
Communities	Events	Green Space	Nature	Music
Infrastructure	Light	People	Visual Design	Spirituality/Religion
Plants	Smells	Sounds	Life & Death	Epiphany
Vistas	Water	Weather		

Phenomenologies

Tensions

<u>Formal</u> Ornate ↔ Functional	<u>Social</u> Inclusive ↔ Exclusive	<u>Ecological</u> Wild ↔ Curated
<u>Evaluative</u> Impressive ↔ Ordinary	<u>Temporal</u> Novel ↔ Familiar	<u>Spatial</u> Expansive ↔ Intimate

Binaries

Alterity	Surprise	Threshold	Sanctuary
----------	----------	-----------	-----------

Temporalities

History		
Ephemeral	Periodic	Recurring
Seasonal	Daily	Timeless

5.3 Activating Awe

Create opportunities to practice awe:

- Promote slowness and mindfulness as preconditions for awe
- Prioritize walkability, since awe is most often noticed at walking pace
- Design for the body, recognizing that awe is an embodied, sensory experience
- Build emotional granularity as a civic capacity through educational campaigns, public language, signage, and field guides that help people name what they are feeling
- Cultivate the skill of noticing awe through educational campaigns, public programming, and resources that train attention toward the everyday rhythms where awe most often surfaces
- Institutionalize the awe walk as a low-cost, scalable, and replicable practice with mental health, prosocial, and place-based benefits

Allow awe to be shared

- Treat residents as co-researchers in the experience of awe so that they can deepen their relationship and familiarity with their communities and neighborhoods
- Recognize that awe is self-reinforcing and wants to be communicated, and that noticing begets noticing
- Host events and small gatherings where awe can be shared

Analyze experiences of awe:

- Approach awe as rhythmic, recognizing that every site has a diverse set of qualities
- Collect data and establish baselines to track how experiences of awe shift over time
- Classify and analyze experiences of awe based on their geographies, morphologies, phenomenologies, and temporalities
- Use frequencies and pairings of variables to identify strengths and weaknesses, treating underproduction and overproduction of awe as indicators
- Analyze awe spatially to reveal hotspots and corresponding anchoring locations
- Analyze awe categorically to surface archetypes and tensions, since each archetype carries its own planning logic and each tension can reveal the kind of intervention a place needs

Align with sacredness

- Practice restraint, recognizing that not every site should be named, mapped, or programmed and that some rhythms must remain unnamed for their mystery to remain intact
- Work to release the potentialities of awe rather than capturing them
- Design sanctuaries deliberately as a building typology, including meditation rooms in public and private institutions, silent corners in parks, and pocket sanctuaries embedded throughout the city
- Design thresholds, including gateways, doorways, and edges that help to mark passage from one register to another
- Treat history as living time through rituals, gatherings, seasonal events
- Find a ‘simple center’ and remove noise and distractions before elaborating
- Design for direct personal encounter rather than mediated experience

Address the awful

- Analyze, map, and identify categories of awfulness, acknowledging that awfulness may diminish or counteract moments of positive awe
- Address awfulness through interventions like traffic-calming, noise abatement, anti-displacement initiatives, and remediation of degraded spaces
- Distinguish actively harmful awfulness from ordinary, neutral spaces that can be left alone

Cultivate the conditions of awe:

- Provide ample opportunities for interconnection and moral beauty as the social and ethical wellsprings of awe in everyday life
- Create spatial conditions across the full range of tensions, including expansiveness and intimacy, ornateness and functionality, novelty and familiarity
- Invest in unnecessarily beautiful things and attention to detail
- Approach awe as an ecological project, since more parks, trees, and green spaces invite the rhythms of wildlife back into the city
- Uphold and re-establish clear spatial boundaries between urban and ecological zones so that nature has space to flourish on its own terms and residents have meaningful thresholds to pass through
- Honor and preserve existing wellsprings of awe—culturally significant places, ecologically rich sites, historic architecture—by limiting encroachment and enforcing preservation requirements even when current trends pull in other directions
- Recognize maintenance as a form of design for the rhythms that produce awe depend on care and renewal to remain perceptible

Embrace an urban ethic:

- See cities as emotional spaces, defined by how we want them to feel rather than only by how they function or what they produce
- Move from designing for awe to designing with awe by building reciprocal relationships with the rhythms and conditions that elicit it
- Take responsibility for the systems, spaces, and communities that constitute urban life so that they retain their capacity for diverse rhythms and their potential for sacredness
- Accept that an urban ethic of awe will mean disrupting the stability and order we've grown accustomed to, and recognize what stands to be gained in return

5.4 The Awe Atlas

As a way to ground the dataset and findings of this research beyond the page, and inspire awe, The Awe Atlas is a public-facing web tool designed ground the dataset and findings of this research beyond the page, and inspire new ways to think about and engage with awe. Hosted at tyler-barron.github.io/awe-cambridge, The Awe Atlas presents the 220 georeferenced instances of awe collected through this study. It provides a story of awe in Cambridge through the archetypes and their spatial clustering, and concludes with a set of tools and resources for interested users and practitioners. Tool 01, “Encounter Awe,” provides a speculative wayfinding application that could be used to find previously georeferenced instances of awe. Tool 02, “Explore Awe,” provides a doorway into the dataset through both an interactive map and an alluvial diagram that together, allow users to better understand the categorical and spatial nature of awe. Tool 03, “Share Awe,” is a speculative tool that provides a space for users in a given location to gather on awe walk, to share moments of awe through awe walks, and that allows people to add new moments of awe to the The Awe Atlas. As a way to guide users, Resource 01, “Understanding Awe,” offers a more robust way to learn about the geographies, morphologies, phenomenologies, and temporalities of awe. Resource 02, “In Awe of Planning Master’s Thesis,” provides a link to this research study and gives users a chance to further their understanding of awe.

The *Awe* Atlas

TOOL 01

Encounter *Awe*

A speculative tool to browse 220 georeferenced instances of awe near you and navigate to a place by compass or AI directions.

TOOL 02

Explore *Awe*

INTERACTIVE MAP

Locate instances

220 georeferenced instances plotted across Cambridge. Filter by morphology, phenomenology, temporality, and more.

ALLUVIAL EXPLORER

Visualize patterns

This is a tool for a deeper dive into the research. Use it to see how instances of awe flow across analytical categories.

TOOL 03

Share *Awe*

A speculative tool to share your own moments of awe to the atlas, find and join upcoming awe walks, and attend or host an awe talk — a small gathering to share what moves you.

RESOURCE

Understanding Awe

A field guide to the dimensions of awe — morphology, phenomenology, temporality, and the categories used throughout these tools.



*References &
Appendices*

References

Allen, S. (2018). *The science of awe* [White paper]. Greater Good Science Center.
https://ggsc.berkeley.edu/images/uploads/GGSC-JTF White Paper-Awe_FINAL.pdf

Anthropic. (2026). *Claude* (Opus 4.7 & Sonnet 4.6) [Large language model]. <https://claude.ai>

Bai, Y., Maruskin, L. A., Chen, S., Gordon, A. M., Stellar, J. E., McNeil, G. D., Peng, K., & Keltner, D. (2017). Awe, the diminished self, and collective engagement: Universals and cultural variations in the small self. *Journal of Personality and Social Psychology*, 113(2), 185–209. <https://doi.org/10.1037/pspa0000087>

Bivand, R. S., Pebesma, E., & Gómez-Rubio, V. (2013). *Applied spatial data analysis with R* (2nd ed.). Springer.
<https://doi.org/10.1007/978-1-4614-7618-4>

Brown, B. (2021). *Atlas of the heart: Mapping meaningful connection and the language of human experience*. Random House.

Büssing, A. (2021). Wondering awe as a perceptive aspect of spirituality and its relation to indicators of wellbeing: Frequency of perception and underlying triggers. *Frontiers in Psychology*, 12, 738770.
<https://doi.org/10.3389/fpsyg.2021.738770>

Clegg, S. R., & Bailey, J. R. (Eds.). (2008). Alterity (otherness). In *International encyclopedia of organization studies* (Vol. 4, p. 55). SAGE Publications. <https://doi.org/10.4135/9781412956246.n19>

Curtin, S. (2009). Wildlife tourism: The intangible, psychological benefits of human–wildlife encounters. *Current Issues in Tourism*, 12(5–6), 451–474. <https://doi.org/10.1080/13683500903042857>

Degen, M. (2010). Consuming urban rhythms: Let's Ravalejar. In T. Edensor (Ed.), *Geographies of rhythm: Nature, place, mobilities and bodies* (pp. 21–32). Ashgate Publishing Company.

Edensor, T. (Ed.). (2010). *Geographies of rhythm: Nature, place, mobilities and bodies*. Ashgate Publishing Company.

Elden, S. (2004). Rhythmanalysis: An introduction. In H. Lefebvre, *Rhythmanalysis: Space, time and everyday life* (S. Elden & G. Moore, Trans., pp. vii–xv). Continuum.

Friedlander, J. (2026). *Take an 'awe walk': The benefits of walking with presence*. Stanford Lifestyle Medicine.
https://lifestylemedicine.stanford.edu/awe_walk_gratitude/

Gehl. (n.d.). Cities of awe. <https://www.citiesofawe.com/>

Jiang, L., Yin, J., Mei, D., Zhu, H., & Zhou, X. (2018). Awe weakens the desire for money. *Journal of Pacific Rim Psychology*, 12, e4. <https://doi.org/10.1017/prp.2017.27>

Kashdan, T. B., Barrett, L. F., & McKnight, P. E. (2015). Unpacking emotion differentiation: Transforming unpleasant experience by perceiving distinctions in negativity. *Current Directions in Psychological Science*, 24(1), 10–16. <https://doi.org/10.1177/0963721414550708>

Keltner, D. (2023). *Awe: The new science of everyday wonder and how it can transform your life*. Penguin Press.

Keltner, D., & Haidt, J. (2003). Approaching awe, a moral, spiritual, and aesthetic emotion. *Cognition and Emotion*, 17(2), 297–314. <https://doi.org/10.1080/026999303022297>

Krasin, K. E. (1972). *Sacred experience and place in cities* [Master's thesis, Massachusetts Institute of Technology]. DSpace@MIT. <https://dspace.mit.edu/bitstream/handle/1721.1/75573/25968857-MIT.pdf>

Lê, S., Josse, J., & Husson, F. (2008). FactoMineR: An R package for multivariate analysis. *Journal of Statistical Software*, 25(1), 1–18. <https://doi.org/10.18637/jss.v025.i01>

Lefebvre, H. (2004). *Rhythmanalysis: Space, time and everyday life* (S. Elden & G. Moore, Trans.). Continuum.

Mateer, T. (2022). Developing connectedness to nature in urban outdoor settings. *Frontiers in Psychology*, 13. <https://doi.org/10.3389/fpsyg.2022.940939>

Monroy, M., & Keltner, D. (2023). Awe as a pathway to mental and physical health. *Perspectives on Psychological Science*, 18(2), 309–320. <https://doi.org/10.1177/17456916221094856>

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23. <https://doi.org/10.2307/2332142>

Piff, P. K., Dietze, P., Feinberg, M., Stancato, D. M., & Keltner, D. (2015). Awe, the small self, and prosocial behavior. *Journal of Personality and Social Psychology*, 108(6), 883–899. <https://doi.org/10.1037/pspi0000018>

R Core Team. (2024). *R: A language and environment for statistical computing* [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>

Rudd, M., Vohs, K. D., & Aaker, J. (2012). Awe expands people's perception of time, alters decision making, and enhances well-being. *Psychological Science*, 23(10), 1130–1136. <https://doi.org/10.1177/0956797612438731>

Saroglou, V., Buxant, C., & Tilquin, J. (2008). Positive emotions as leading to religion and spirituality. *The Journal of Positive Psychology*, 3, 165–173. <https://doi.org/10.1080/17439760801998737>

References & Appendices

Stellar, J. E., John-Henderson, N., Anderson, C. L., Gordon, A. M., McNeil, G. D., & Keltner, D. (2015). Positive affect and markers of inflammation: Discrete positive emotions predict lower levels of inflammatory cytokines. *Emotion*, 15(2), 129–133. <https://doi.org/10.1037/em00000033>

Sturm, V. E., Datta, S., Roy, A. R. K., Sible, I. J., Kosik, E. L., Veziris, C. R., Chow, T. E., Morris, N. A., Neuhaus, J., Kramer, J. H., Miller, B. L., Holley, S. R., & Keltner, D. (2022). Big smile, small self: Awe walks promote prosocial positive emotions in older adults. *Emotion*, 22(5), 1044–1058. <https://doi.org/10.1037/em00000876>

The Aldo Leopold Foundation. (n.d.). *Understanding the land ethic*. Retrieved May 11, 2026, from <https://form.jotform.com/252226392191152>

Tugade, M. M., & Fredrickson, B. L. (2004). Resilient individuals use positive emotions to bounce back from negative emotional experiences. *Journal of Personality and Social Psychology*, 86(2), 320–333. <https://doi.org/10.1037/0022-3514.86.2.320>

Valdesolo, P., & Graham, J. (2014). Awe, uncertainty, and agency detection. *Psychological Science*, 25(1), 170–178. <https://doi.org/10.1177/0956797613501884>

Van Cappellen, P., & Saroglou, V. (2012). Awe activates religious and spiritual feelings and behavioral intentions. *Psychology of Religion and Spirituality*, 4(3), 223–236. <https://doi.org/10.1037/a0025986>

Yaden, D. B., Kaufman, S. B., Hyde, E., Chirico, A., Gaggioli, A., Zhang, J. W., & Keltner, D. (2019). The development of the Awe Experience Scale (AWE-S): A multifactorial measure for a complex emotion. *The Journal of Positive Psychology*, 14(4), 474–488. <https://doi.org/10.1080/17439760.2018.1484940>

Initial Interview Protocol

Relationship to Cambridge and Awe/Non-awe Group:

- 1. How long have you been in Cambridge, and what brought you here?
- 2. Can you tell me a bit about your awe-related/non awe-related work in Cambridge—what drew you to it, and what's your typical experience like

Use of Public Spaces:

- 3. When you think about your daily or weekly routines in Cambridge, what public spaces do you move through or spend time in most often?

Connection to Systems:

- 4. Are there things that help you to feel like you're a part of Cambridge? If yes, what are those things? (e.g. relationships, communities, networks, traditions, natural cycles, cultural groups, etc)

Experiences of Awe:

- 5. Can you tell me about a specific moment in Cambridge when you've felt a sense of wonder?
- 6. Are there specific moments in Cambridge when you felt part of something vast or larger than yourself?
- 7. Are there specific moments when you've felt interconnected or interdependent with others?
- 8. Are there any moments when you felt a sense of mystery?

Spatial Qualities Associated with Awe:

- 9. How would you describe the quality of the spaces in which you've experienced a sense of wonder, interdependence or mystery?

Attention and Embodiment of Awe: perceptual habits, and disruption of routines

- 10. When you're moving through Cambridge, what catches your attention—what makes you pause, appreciate things, or change your usual routine?

Follow-up Interview Protocol

Post-Conversation Reflection:

- 1. Did you notice anything shift for you after our initial conversation—in how you moved through Cambridge or what you paid attention to?
- 2. Can you describe a specific moment or place where you experienced awe during this period of time?
- 3. Did the types or frequency of awe experiences change for you over these weeks?
- 4. Has your relationship to Cambridge shifted?
 - a. If yes, how?
- 5. Has your awareness of awe or public spaces shifted?
 - a. What have you noticed that you haven't noticed before?
 - b. Have you changed any behaviors?
- 6. Did you discover anything new—either new places, or something new about familiar places?

Awe Walk:

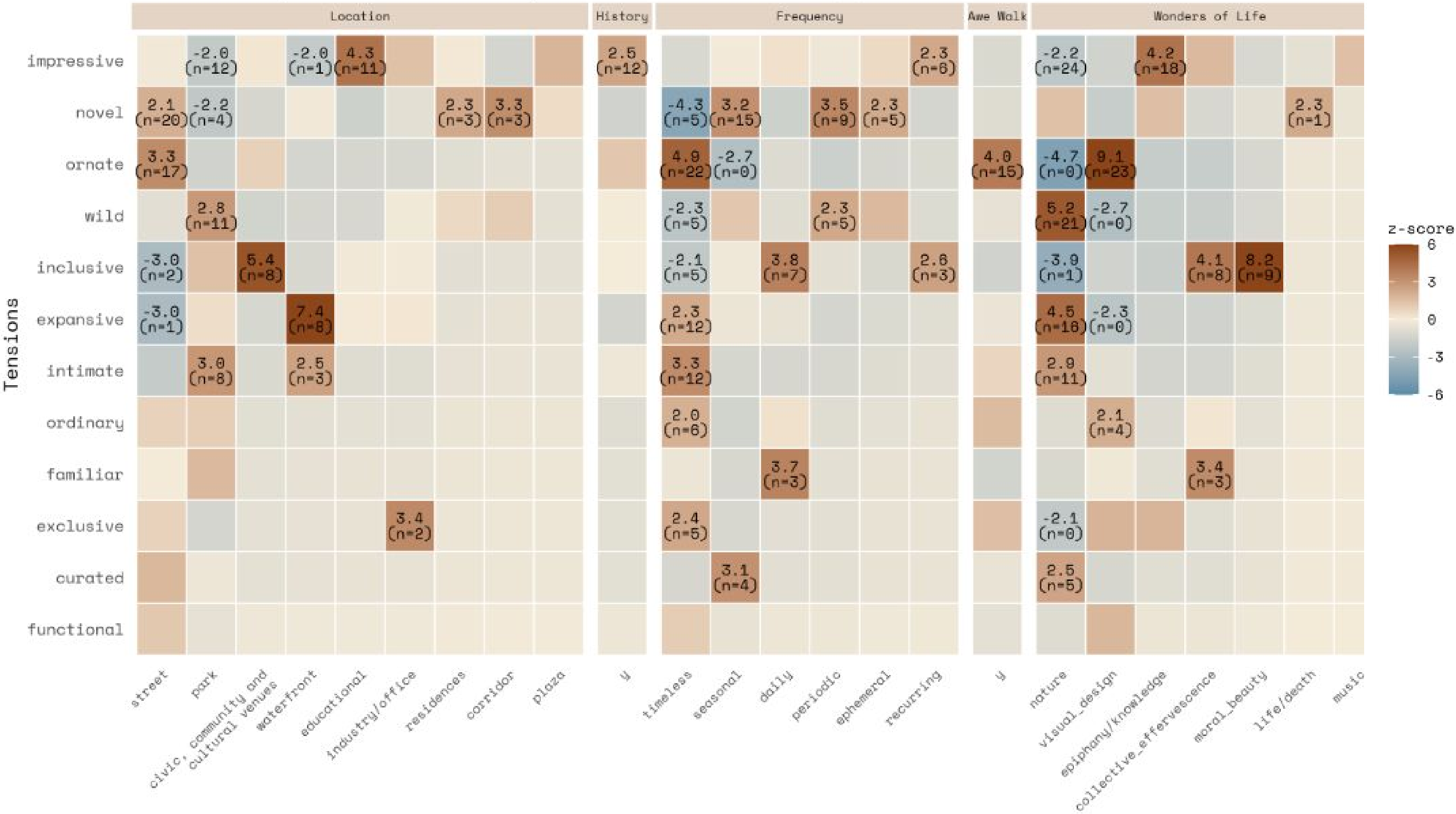
- 7. What was it like to do an awe walk in Cambridge? Where did you go? How long? What stood out to you? (Can you describe the start and end points or the general route)
 - a. Are there any places you thought to avoid?
- 8. What kinds of awe did you experience during your walk? (i.e moments of wonder, vastness, mystery, feeling small, interconnection)
- 9. What was the quality of the space(s) where you experienced awe during your walk (i.e., Was it green and filled with nature, were there other people walking, kids playing, etc)?

Future Practice:

- 10. Do you think the awe walk, or this study will have any lasting impact on how you relate to Cambridge, your neighborhood, a certain place, or the work your group does? If so, what will change in the future?

Residual Matrix: Tensions vs. all other variables

Brown = over-represented; blue = under-represented; cells labeled where |z| > 2



Appendix B. Residual Matrix, Part 2

Residual Matrix: Tensions vs. all other variables

Brown = over-represented; blue = under-represented; cells labeled where |z| > 2

